

# Peak Age of Information over Scheduled, Energy-Constrained Multi-Segment Delay-Tolerant Networks: Bundle Replication under Route Uncertainty

Haoyu Wang<sup>a</sup>, Bo Sheng<sup>a</sup>, Xiaoqian Zhang<sup>\*b</sup>

<sup>a</sup>Department of Computer Science, University of Massachusetts Boston, Boston, MA 02125, USA

<sup>b</sup>Department of Computer Science, University of Nebraska at Omaha, Omaha, NE 68182, USA

## Abstract

We study information freshness—the age of information and its peak—over scheduled, intermittently connected, multi-segment delay-tolerant networking paths spanning low-Earth-orbit satellite, UAV, and terrestrial segments, in which bundles are replicated across contact opportunities under an energy-harvesting budget. Existing freshness analyses address persistent links with random failures, and recent satellite studies treat intermittent store-carry-forward connectivity; no prior work jointly couples deterministic contact-plan gating, multi-copy replication, and an energy budget. We derive closed-form residual-wait and peak-age laws for scheduled contacts; a phase-mixing composition rule whose two-copy gain equals half the mean absolute dispersion of path latencies and operates precisely under route-outcome uncertainty; and an energy–replication threshold: the peak-age-optimal copy degree brackets the energy harvested per contact period, extends to per-copy route success, and yields a constant-time policy on top of unmodified Contact Graph Routing. All results are verified by model-exact Monte-Carlo, a Bundle Protocol/CGR simulator, real Iridium-NEXT ephemeris, and the ION reference implementation.

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**KEYWORDS:** Age of information, peak age of information, delay-tolerant networking, contact graph routing, LEO satellites, bundle replication, energy harvesting

## 1. Introduction

Sixth-generation networks are expected to integrate non-terrestrial segments—LEO mega-constellations, high-altitude platforms, and UAVs—to extend connectivity beyond terrestrial reach [42, 43]. Many of these links are not persistently available: a ground station sees a given LEO satellite only during short, periodic passes [44], a UAV revisits a sensor field on a semi-regular patrol, and opportunistic terrestrial contacts come and go at random. Connectivity is therefore *scheduled* and *intermittent*, and is handled by the delay-tolerant networking (DTN) architecture: data are carried as bundles, stored and forwarded across predicted contact opportunities, and routed by Contact Graph Routing (CGR) over a contact plan.

For an important class of traffic—telemetry, situational awareness, disaster response, and IoT monitoring—the figure of merit is not throughput or delivery ratio but *information freshness*, captured by the Age of Information (AoI) and its peak (PAoI). Freshness over such paths is shaped by two effects that the DTN setting makes unavoidable. First, a bundle generated mid-gap must *age in a buffer* until the next scheduled contact, so the freshness penalty is governed by a *residual-wait* law

set by the contact schedule, not by a queueing delay on an always-on link. Second, the standard DTN tool for combating uncertain contacts is *replication*—sending multiple bundle copies over different contacts—but on power-limited, energy-harvesting satellites and UAVs each copy costs energy, so replication and freshness are coupled through an energy budget.

These ingredients exist in largely separate literatures (the nearest junction, joint sampling/routing under intermittent links and energy, is distinguished in Section 2). Freshness analyses for satellite and multi-hop networks largely model persistent servers with random erasure or randomly failing relays, with recent exceptions treating intermittent satellite connectivity and store-and-forward delivery (Section 2); tandem-queue AoI results assume persistent service; and the space-DTN/contact-plan community optimizes delivery ratio under uncertain contacts without a freshness or energy objective. None couples *scheduled store-carry-forward gating*, *multi-copy replication*, and an *energy constraint*. We close this gap with a tractable PAoI framework spanning the three segment types, an end-to-end composition rule with an explicit independence condition, and an energy–replication threshold that says when spending energy on extra copies actually buys freshness. The precise problem is *not* “replication improves freshness on scheduled paths”—with no uncertainty, an oracle single-copy router ties replication (Remark 8)—but: given deterministic scheduling as the *structure* and route-outcome *uncertainty* (unknown phases, relay faults, plan er-

<sup>\*</sup>Xiaoqian Zhang (Corresponding author), Department of Computer Science, University of Nebraska at Omaha, Omaha, NE 68182, USA (email: xiaoqianzhang@unomaha.edu).

ror) as the *source* of the gain, how many copies does a harvesting energy budget justify, and where should they be placed? We deliberately build on, rather than compete with, the CGR/RUCoP contact-plan lineage.

Several constituent identities below are classical—the residual law is inspection-paradox/renewal theory specialized to duty-cycled contacts, and the two-copy gain is a Gini-mean-difference identity; we state them as anchors, not as contributions. The contribution is the *joint construction*: scheduled contact-plan freshness, explicit phase structure (ensemble versus fixed plan), energy-coupled replication degree under uncertainty, and its realization in the BP/CGR stack. Our contributions are:

- A closed-form LEO residual-wait law and AoI/PAoI anchor (Section 4), generalized to UAV (semi-periodic) and terrestrial (renewal) segments by one alternating-renewal model, which also yields an exact AoI/PAoI *inversion* condition (Section 4.1);
- A phase-mixing independence lemma and a  $k=2$  PAoI composition theorem showing the replication gain equals half the mean absolute dispersion of path latencies (Section 5)—a gain operative exactly under route-outcome uncertainty (Remark 8);
- An energy–replication threshold: a conservative no-starvation degree  $\lfloor \lambda_e P/e \rfloor$  and a skip-aware PAoI optimizer  $k^*$  at the adjacent integer, capped at  $K_{\max}$ , proved monotone in the harvest-to-contact rate and saturating once energy ceases to bind—stated for the phase-agnostic (random-offset) gain profile and shown to carry over to deterministically staggered fixed plans, whose exact gain follows from the joint residual survival computable from the contact plan—and hardened by an exact work-conserving battery-queue throttle (Sections 6–7);
- A concrete AoI-Energy replication policy—a local, closed-form two-candidate rule (test only  $\lfloor \eta \rfloor$  and  $\lceil \eta \rceil$ ,  $O(1)$ ) that sets the per-update copy count from the node’s mean energy-adequacy on top of CGR routing—which matched the exhaustive finite-battery PAoI optimum at every tested i.i.d.-harvest  $(\eta, B)$  configuration (and whose measured failure regime under strongly persistent harvest we report), dominates any fixed degree across the evaluated i.i.d.-harvest energy range, and (against a RUCoP-style MDP core we implemented) attains most of its delivery at a fraction of the energy on the illustrative instance tested (Section 8);
- *Phase-aware copy placement*: on real Iridium-NEXT ephemeris the route-set *latency-diversity potential*—the residual-min ratio  $\mathbb{E}[R_{\min,2}]/\mathbb{E}[R_1]$ —spans 0.41–0.95 with the pass-phase alignment of the chosen satellite pair. Under route uncertainty this diversity is what replication converts into freshness (without uncertainty, a plan-aware single-copy router exploits the same diversity; Remark 8); because contact plans are known in advance, the alignment is computable at planning time, turning the independence assumption into a route-selection criterion (Section 9);
- Model-exact Monte-Carlo validation of every result, corroboration in a DTN-realistic CGR simulator, a cross-check of the residual-wait anchor in the ION reference implementation, with code, scenarios, and result data public (data-availability statement; Section 9).

## 2. Related Work and Positioning

*Age of Information: metric and foundations.* The AoI metric was introduced by Kaul *et al.* [1] to capture the freshness of status updates; peak AoI and packet-management effects were formalized by Costa *et al.* [2], and the area is surveyed in [3, 4], with the broader “X of information” metric family (age, value, semantics) reviewed in [5]. PAoI optimization and its distribution have since been studied for single- and multi-class queues [6, 7]. These works establish AoI/PAoI as the freshness metric; we carry them into the scheduled, gated, multi-segment DTN setting they do not treat.

*AoI in queues, tandems, and multi-hop paths.* Multi-hop AoI has been characterized through age-optimal scheduling [8], stochastic-hybrid-system moment methods [9], and freshness-optimal link scheduling under interference [10]; Kam *et al.* [11] show that *path (replica) diversity* lowers status age—the queueing antecedent of our min-over-copies replication gain. Recursive PAoI in a tandem of bufferless servers [12] (whose recursion we borrow for general- $k$  numerics) and unreliable tandems with random *server breakdown* [13] are the closest queueing models, but all assume persistent—or memorylessly failing—service; a contact plan is scheduled deterministic-periodic, giving a fundamentally different residual-wait law ((5); memoryless breakdown is one point on our CV axis).

*AoI over unreliable and erasure channels.* Freshness over erasure/noisy links has been analyzed with coding and retransmission [14, 15], including peak-age violation probability under short-packet HARQ [16] and random access [17]; Chiariotti *et al.* [18] give average-AoI and PAoI-distribution bounds for multi-hop tandem connections (LEO-motivated in the companion preprint) with packet erasure and tributary traffic. All assume an *always-on* server subject to random erasure. We differ in that the server is OFF for long *scheduled* intervals during which bundles age in buffer—and we add replication and an energy budget; our tail/deadline analysis is in the spirit of [16] but over scheduled gating.

*Freshness under energy constraints.* A large body of work minimizes AoI at an energy-harvesting source: offline and online policies under energy replenishment [19, 20], threshold-optimal transmission with finite batteries [21], the age–energy tradeoff showing that greedy updating becomes suboptimal once harvested energy falls below demand [22] (an anchor for our threshold), the general update-or-wait principle [23], and learning-based control under HARQ and sensing cost [24]; resource-aware, learning-driven control is likewise maturing in adjacent networked-control settings such as vehicular platooning [25]. These all decide *when* a single source transmits; none couples the energy budget to the *replication degree* over scheduled multi-hop contacts, which is our central object.

*DTN architecture, Bundle Protocol, and Contact Graph Routing.* Our setting is the DTN architecture [26, 27] and its Bundle Protocol [28, 29], routed by Contact Graph Routing over a contact plan [30] with a reference implementation in ION [31]. Scheduled connectivity is captured by contact-plan design [32, 33], and CGR performs well across satellite constellations [34]. Routing under *uncertain* contact plans is cast as a multiple-copy MDP in the RUCoP/CGR-UCoP family [35], while CGR-ETO and overbooking management [36] refine contact scheduling. Energy enters this lineage explicitly: E-CGR [37] validates and confirms CGR routes against satellite battery and harvesting constraints, and 3DEHR [38] routes around energy-poor nanosatellites via an energy-harvesting-aware potential field. These works optimize delivery ratio or delay—subject, in the energy-aware variants, to energy *feasibility*—but none optimizes *freshness*: our delta is the freshness-aware replication *degree* under scheduled contacts, i.e. how many copies an energy budget is worth in PAoI terms. We adopt the multi-copy, contact-plan framing so our results compose with—rather than replace—the CGR stack (Section 8).

*Replication and opportunistic routing.* Copy-based DTN routing spans uncontrolled flooding [39], bounded-copy Spray-and-Wait [40], and delivery-predictability forwarding [41]. These fix or heuristically bound the copy count; we instead *derive* the PAoI-optimal, energy-aware degree and its threshold; copy-count heuristics designed for unknown contact opportunities are not natural baselines in a known-contact-plan setting, so our comparisons are against plan-aware alternatives (Section 8).

*Non-terrestrial networks, SAGIN, and UAV mules.* 6G integrates non-terrestrial segments [47, 48] into space-air-ground networks [49], in which UAVs serve as data mules or message ferries [50, 51], their revisit patterns set by trajectory-planning frameworks such as [52] (the UAV-network design space is surveyed in [46])—precisely the schedule our semi-periodic UAV segment abstracts. Freshness in these set-

**Table 1**

Positioning against the closest prior art.

| Work                | scheduled<br>gating | multi-copy<br>replication | energy<br>budget | freshness<br>metric |
|---------------------|---------------------|---------------------------|------------------|---------------------|
| Chiariotti [18]     | —                   | —                         | —                | AoI/PAoI            |
| Ari [56]            | ✓                   | —                         | —                | PAoI                |
| Sinha [12]          | —                   | —                         | —                | PAoI                |
| Senthilkumar [13]   | rand.               | —                         | —                | AoI                 |
| RUCoP/CGR-UCoP [35] | ✓                   | ✓                         | —                | —                   |
| Isayama [54]        | —                   | —                         | —                | AoI                 |
| E3F [55]            | —                   | ✓                         | ✓                | —                   |
| Atasayar [61]       | rand.               | —                         | ✓                | AoI                 |
| <b>This work</b>    | ✓                   | ✓                         | ✓                | AoI/PAoI            |

tings has been studied for UAV-assisted IoT (peak-AoI-optimal trajectories) [53], disaster DTN [54], energy-aware emergency frameworks [55], and—closest to us—AoI under intermittent satellite coverage and store-and-forward [56], finite-horizon scheduling over intermittent LEO coverage [57], and energy-harvesting sources feeding LEO constellations [58]. Concurrent work by Zhao *et al.* [60] couples LEO intermittent ON/OFF connectivity to an energy buffer via semi-Markov methods with probe-before-transmission control; unlike ours, it does not treat multi-copy bundle replication, contact-plan-aware route placement, or CGR. Closest in scope, Atasayar *et al.* [61] jointly optimize sampling and routing for nonlinear AoI over intermittent links with *correlated* route delays and route-dependent energy costs, via a constrained semi-Markov decision process—the nearest prior junction of intermittency, routing, and energy. The delta is structural: their links are *stochastically* available and the decision is which single route to sample onto, whereas ours are *deterministically scheduled* contact plans (yielding closed-form residual laws rather than an MDP), the control is the bundle *replication degree* against a harvesting battery, and the result is realized in the BP/CGR stack. Most directly on the replication axis, Badia *et al.* [59] show that spreading updates across DTN relays reduces AoI under intermittent connectivity, obtaining an averaging gain from copy diversity. That work, like [11], treats copy diversity under generic/random intermittency and is energy-agnostic; our contribution is the complementary regime—*deterministic scheduled* contacts (a duty-cycled contact plan, not random availability), a finite-battery copy-degree control with the energy–replication threshold it induces, and a CGR realization—so the gain is characterized against a per-segment residual-wait law rather than an i.i.d. availability model. None of these combines a heterogeneous multi-segment scheduled path, bundle replication, and an energy constraint—the four axes of Table 1.

Table 1 summarizes the positioning along the four axes that define our setting.

### 3. System Model

A status source delivers updates to a remote monitor over an ordered chain of  $S$  store-carry-forward segments, indexed  $s = 1, \dots, S$ . In the canonical instance (Fig. 1), a ground sensor reaches a ground gateway through a UAV mule and a LEO relay—the sensor acting as the source and the gateway as the monitor—so the path is ground sensor  $\rightarrow$  UAV  $\rightarrow$  LEO  $\rightarrow$  ground gateway. Each segment is a server whose availability is dictated by a contact schedule rather than being continuous.

*Per-segment contact abstraction.*

- **LEO:** deterministic periodic ON/OFF, window  $W_s$  once per period  $P_s$  (Section 4);
- **UAV:** semi-periodic / jittered renewal ON/OFF (reduces to the LEO case as jitter  $\rightarrow 0$ ; genuinely Markov-modulated schedules, with correlated successive gaps, are an extension);
- **Terrestrial:** stochastic renewal ON/OFF.

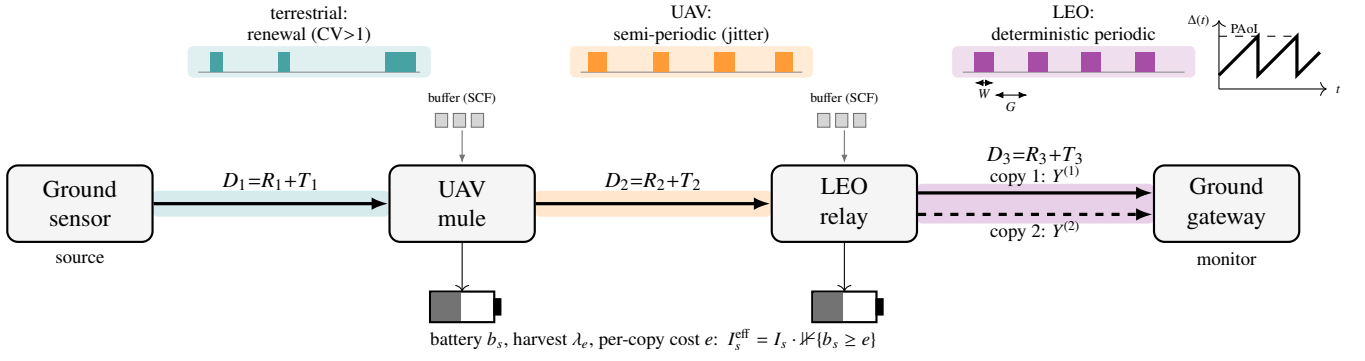
*Quantities and the freshness metrics.*  $\Delta(t) = t - g^*(t)$  is the AoI at the monitor, where  $g^*(t)$  is the generation time of the freshest *delivered* bundle; it drifts up at unit rate and resets whenever a delivery brings fresher information (a stale or duplicate copy causes no reset). Intermittent gating supports two distinct traffic models, and their peak-age processes differ—so we define *three* peak quantities and state, for every result, which one it characterizes (Fig. 2):

- **per-reset PAoI** (textbook): the average of the age just before *every* reset. When a window delivers a back-to-back burst of updates it also counts the small intra-burst peaks, deflating the average;
- **outage-peak age OPAoI:** the average, over contact cycles, of the maximum age just before that contact’s deliveries—the worst-case staleness per silent gap. Under *generate-at-will* traffic (fresh data available at every window) the single-LEO cycle peak is OPAoI =  $T_s + G$  (Section 4);
- **per-update PAoI<sup>per</sup>:** the standard PAoI when the source launches one update per firing epoch (paced generation): each update yields one reset, and  $\mathbb{E}[\text{PAoI}^{\text{per}}] = \mathbb{E}[Z] + \mathbb{E}[Y_{\min}]$  with  $Z$  the inter-firing interval and  $Y_{\min}$  the delivered latency. Here per-reset and per-update coincide. The identity presumes deliveries *preserve generation order*: if a newer update overtakes an older one, the older delivery arrives stale and resets nothing. Order preservation is guaranteed whenever the latency spread is bounded by the firing interval,  $Y_{\max} - Y_{\min} \leq Z_{\min}$  (with lattice-supported  $Z$  and bounded  $Y$  this is checkable from the plan; note  $\sum_s G_s$  can exceed the firing interval on long multi-segment paths, where the condition binds). When it fails, PAoI<sup>per</sup> must be evaluated over the subsequence of *freshness-resetting* deliveries, and  $\mathbb{E}[Z] + \mathbb{E}[Y_{\min}]$  no longer generally characterizes or bounds that process: consecutive resets can sit several firing intervals apart while the resetting updates are the atypically fast ones, so the true peak can land on either side (e.g. deterministic  $Z=1$  with repeating latencies 4, 3, 2, 0, 5 resets once per five epochs with peak 5  $> \mathbb{E}[Z] + \mathbb{E}[Y] = 3.8$ ).

These are genuinely different processes: for  $P=1$ ,  $W=0.1$ ,  $T_s \approx 0$ , OPAoI = 0.9 while  $\mathbb{E}[\text{PAoI}^{\text{per}}] = 1.405$  ( $= P + \mathbb{E}[R]$ )—neither is an approximation of 1 and the renewal generalization characterize OPAoI (and time-average AoI); Results 2–3, the tail corollary, and the policy characterize PAoI<sup>per</sup> under paced generation. Our Tier-2 reconstruction clusters resets within one contact window and keeps the burst maximum, i.e. it measures OPAoI (Section 9); the Tier-1 Result-3 simulations measure PAoI<sup>per</sup>. For a single copy, the per-segment latency increment is  $D_s = R_s + T_s$  with  $R_s$  the residual wait and  $T_s$  the lumped (near-deterministic) service plus propagation, and the end-to-end latency is  $Y = \sum_{s=1}^S D_s$ ; with replication degree  $k$  the delivered latency is  $Y_{\min} = \min(Y^{(1)}, \dots, Y^{(k)})$ .

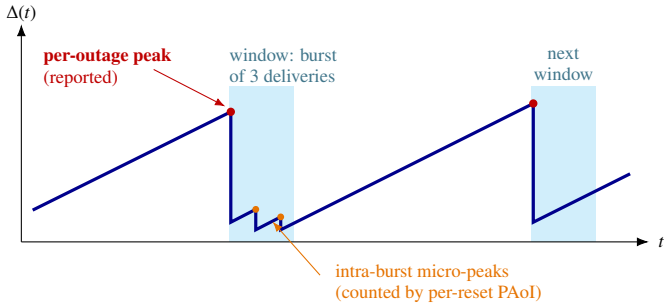
*Uncongested-contact regime (scope).* Throughout, a contact is usable the instant it opens and  $T_s$  absorbs transmission and propagation: we work in the *low-load* regime where contact capacity (window duration  $\times$  rate) exceeds the offered bundle volume, so residual waits—hours—dominate service times—seconds—and replication’s second cost, *contact volume*, is negligible next to its energy cost. Status updates sit deep in this regime (our Iridium scenario uses  $\sim 10^{-5}$  of a window per update), but the assumption is load-bearing: near contact capacity, in-window queueing, overbooking [36], and the volume of extra copies re-enter. Section 9 measures the boundary.

*Energy coupling.* Battery  $b_s(t)$ , harvest rate  $\lambda_{e,s}$ , per-copy energy  $e$ . The effective availability is  $I_s^{\text{eff}}(t) = I_s(t) \cdot \mathbb{1}\{b_s(t) \geq e\}$ : energy scarcity thins contacts (stretches the residual wait) and replication draws the battery down faster (starving future contacts). This tension is resolved in Sections 6–7. *Scope:* although the model admits a battery per segment, the threshold analysis of Sections 6–7 adopts a **single energy-bottleneck abstraction**—one battery (the source-side or tightest node) gates copy launches at constant per-copy cost  $e$ , the binding constraint in our scenarios. Route-dependent costs and interacting per-relay batteries are an



Replication:  $k$  copies over diverse contacts (independent-offset or staggered; §6) (seg 3 shows  $k=2$ ); end-to-end latency  $Y = \sum_s D_s$ , delivered age  $Y_{\min} = \min_k Y^{(k)}$ .

**Fig. 1.** System model. A status source (ground sensor) reaches the monitor (ground gateway) over an ordered chain of *scheduled, intermittently available* segments—terrestrial opportunistic, UAV mule, LEO relay—each a store-carry-forward server (bundles wait in a buffer) that is ON only during contact windows. The color-coded per-segment ON/OFF pattern differs by segment type (high-variance renewal, semi-periodic with jitter, deterministic periodic), so a bundle generated mid-gap ages for a residual wait  $R_s$  before a service+propagation  $T_s$ , giving increment  $D_s = R_s + T_s$  and end-to-end latency  $Y = \sum_s D_s$ . Bundles are *replicated* over  $k$  diverse contacts— independent-offset or deterministically staggered, two distinct regimes treated separately in Section 6 (independent offsets give independent residuals; staggered fixed plans give negatively dependent ones; distinct router next-hops guarantee neither). Segment 3 illustrates  $k=2$  with copies  $Y^{(1)}, Y^{(2)}$  and delivered age  $Y_{\min} = \min_k Y^{(k)}$ . A finite energy-harvesting battery at each relay gates its usable contacts,  $I_s^{\text{eff}} = I_s \cdot \mathbb{1}\{b_s \geq e\}$ . The monitor tracks the Age of Information  $\Delta(t)$  and its peak (PAoI).



**Fig. 2.** The peak-age taxonomy. A contact window delivers a back-to-back burst of distinct updates; each fresher delivery resets the age. The textbook *per-reset* PAoI also averages the small intra-burst peaks, deflating the mean; the *outage peak* OPAoI keeps only the burst maximum—the conservative cycle-level statistic (within each cycle it dominates every intra-burst peak; with variable burst sizes the two averages need not order). Under paced generation (one update per firing epoch) each update yields one reset and the per-update peak PAoI<sup>per</sup> =  $Z + Y_{\min}$  applies; the three coincide when each contact delivers exactly one update.

**Table 2**  
Notation.

|                            |                                                                          |
|----------------------------|--------------------------------------------------------------------------|
| $P, W, G$                  | contact period, window, silent gap $G = P - W$                           |
| $\delta$                   | duty cycle $W/P$ ; $\delta_{\text{eff}} = \delta p_e$ usable-ON fraction |
| $U$                        | arrival phase (Sec. 4); ON duration (Sec. 4.1)                           |
| $V, C$                     | OFF gap, renewal cycle $C = U + V$ (Sec. 4.1)                            |
| $R_s, T_s$                 | residual wait; service+propagation at segment $s$                        |
| $D_s, Y$                   | increment $R_s + T_s$ ; end-to-end latency $\sum_s D_s$                  |
| $k, Y_{\min}$              | replication degree; delivered latency $\min_k Y^{(k)}$                   |
| $Z, \Delta$                | inter-firing interval; age at the monitor                                |
| OPAoI, PAoI <sup>per</sup> | outage-peak age; per-update peak age                                     |
| $b_s, B, e$                | battery level, capacity, per-copy energy cost                            |
| $\lambda_e, \eta$          | harvest rate; energy adequacy $\lambda_e P/e$                            |
| $p_e, L$                   | usable-window probability; cap-overflow loss                             |
| $\mathcal{A}(k), k^*$      | mean-PAoI <sup>per</sup> objective (16); optimal degree                  |
| $K_{\max}$                 | candidate copy routes / contact opportunities                            |

a mixed law with atom  $\mathbb{P}(R = 0) = W/P$  and uniform slab  $f_R(r) = 1/P$  on  $(0, G]$ ;  $\mathbb{P}(R > r) = (G - r)/P$  on  $[0, G]$ ,  $R_{\max} = G$ .

**Theorem 1** (LEO AoI/OPAoI anchor). *With service+propagation  $T_s$ ,*

$$\mathbb{E}[R] = \frac{(P - W)^2}{2P} = \frac{p}{2}(1 - \delta)^2, \quad \mathbb{E}[R^2] = \frac{G^3}{3P}, \quad (2)$$

$$\bar{\Delta}_{\text{LEO}} = T_s + \frac{p}{2}(1 - \delta)^2, \quad \text{OPAoI}_{\text{LEO}} = T_s + P(1 - \delta). \quad (3)$$

The law itself is classical residual-life theory, stated as the anchor on which the DTN-specific structure builds. Note the provisioning consequence: sparse contacts hurt the mean age *quadratically* in  $(1 - \delta)$  but the peak only *linearly*, so mean- and peak-optimal contact provisioning differ.

*Energy-thinning bridge.* If each window is usable with probability  $p_e$  (battery  $\geq e$ ), i.i.d. across periods, the usable schedule is again alternating-renewal with OFF gap  $V' = (J-1)P + G$ ,  $J \sim \text{Geom}(p_e)$ , so every closed form extends through the renewal law of Section 4.1 with  $V \rightarrow V'$ ; in particular the exact thinned residual is

$$\mathbb{E}[R_{\text{eff}}] = \underbrace{\frac{(P - W)^2}{2P}}_{\text{contact gating}} + \underbrace{P \frac{1 - p_e}{p_e}}_{\text{energy thinning}} - \underbrace{\frac{(1 - p_e)W^2}{2P}}_{O(\delta^2)}. \quad (4)$$

We caution that this is a *thinning*, not a rescaling: the shorthand  $\delta_{\text{eff}} = \delta p_e$  preserves only the long-run usable-ON fraction, and substituting  $\delta \mapsto \delta p_e$  into the Theorem 1 forms does *not* reproduce (4)—skipped windows cluster the silence into whole periods, which the

extension we do not claim (Section 9 gates energy at the source accordingly).

**Remark 1** (Report both metrics). *Exact average AoI in tandems is closed-form only to two–three hops, while the peak metrics remain reachable; and under intermittency OPAoI can fall below time-average AoI (Remark 2), so we report peak and average throughout.*

#### 4. Result 1: Residual-Wait Laws for Scheduled Contacts

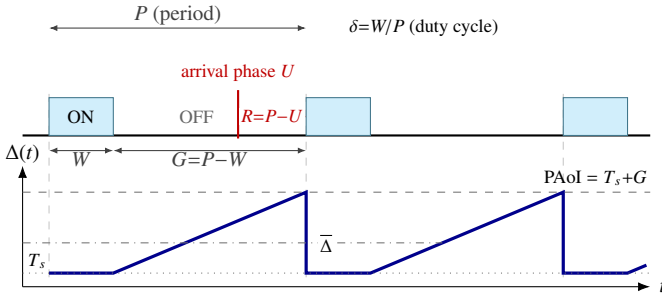
A LEO satellite rises over the ground terminal once per orbital period  $P$  and stays in view for a contact window of duration  $W$ ; the link is therefore ON during  $[0, W]$  of each period and OFF (dark) during the silent gap  $G \triangleq P - W$  that follows, so the terminal is in contact only a fraction  $\delta \triangleq W/P$  of the time—the duty cycle (Fig. 3). A status update generated at a stationary (random-phase) instant meets this schedule at a uniformly random phase  $U \sim \text{Unif}[0, P]$  (justified by Lemma 1): if  $U$  falls inside a window the bundle is forwarded at once, otherwise it must age in buffer until the next pass. This residual wait is

$$R = \begin{cases} 0, & U \in [0, W] \\ P - U, & U \in [W, P] \end{cases} \quad (1)$$

**Table 3**

Modeling assumptions and their failure modes.

| Assumption                         | Where                   | If violated                                                  |
|------------------------------------|-------------------------|--------------------------------------------------------------|
| Decorrelated contacts per copy     | Results 2–3             | same-bottleneck copies: gain $\rightarrow 0$                 |
| Random offsets / spread-out delays | Lem. 1, Rem. 3          | commensurate or lattice: locking                             |
| $k \leq 2$ closed form             | Result 2                | general $k$ via $\int (1 - F_Y)^k$ (numerical)               |
| Generate-at-will                   | Result 1                | finite $\lambda$ : $+1/\lambda$ floor (Sec. 4.1)             |
| Large battery                      | Result 3 (fluid $p_e$ ) | finite $B$ : $p_e \leq \min(1, \eta/k)$ , presses $k^*$ down |



**Fig. 3.** Result 1 parameters. *Top:* the LEO segment is ON for a window  $W$  once per period  $P$  (silent gap  $G=P-W$ , duty cycle  $\delta=W/P$ ); a bundle generated at arrival phase  $U$  in the gap waits a residual  $R=P-U$  for the next window. *Bottom:* the resulting age  $\Delta(t)$  is pinned near the service+propagation floor  $T_s$  during a window and ramps over the gap, peaking at  $\text{OPAol}=T_s+G$ ; the time-average is  $\Delta=T_s+(P-W)^2/2P$  (Theorem 1).

residual's second-moment dependence ( $\mathbb{E}[V^2]/2\mathbb{E}[C']$ ) captures and a duty-cycle rescaling does not. The i.i.d. premise itself is a mean-field approximation—a finite battery correlates consecutive windows—made precise in Section 7.

#### 4.1. Renewal Generalization: UAV and Terrestrial Segments

Real UAV and terrestrial links are not perfectly periodic: a UAV mule revisits the sensor field on a jittered patrol, and opportunistic terrestrial contacts appear at random. One *alternating-renewal* model captures all three segment types—with the deterministic LEO schedule as a special case—by letting each contact cycle be an ON period of duration  $U$  (the window) followed by an OFF gap of duration  $V$  (mean  $G$ , the dark interval before the next contact), where  $U$  and  $V$  may now be random; the cycle length is  $C = U + V$  and the duty cycle is  $\delta = \mathbb{E}[U]/\mathbb{E}[C]$ . By renewal-reward and the inspection paradox, a stationary arrival sees

$$\Pr(R=0) = \delta, \quad f_R(r) = \frac{\Pr(V > r)}{\mathbb{E}[C]}, \quad (5)$$

$$\mathbb{E}[R] = \frac{\mathbb{E}[V^2]}{2\mathbb{E}[C]}, \quad \mathbb{E}[\text{OPAol}] = T_s + \mathbb{E}[V].$$

Deterministic  $V$  recovers Theorem 1 (LEO); a low-variance  $V$  models UAV semi-periodic jitter ( $\rightarrow$  LEO as the jitter vanishes); exponential  $V$  is the 2-state CTMC (the “unreliable-server” case, here a special case rather than the contribution); heavy-tailed  $V$  models opportunistic terrestrial contacts.

**Remark 2** (AoI/PAoI inversion). Mean OPAoI depends only on  $\mathbb{E}[V]$  while mean AoI grows with the gap variance, so  $\mathbb{E}[\text{OPAol}] < \Delta \iff \mathbb{E}[V^2] > 2\mathbb{E}[C]\mathbb{E}[V]$ . Writing  $C = U + V$ , this is exactly

$$\text{CV}^2(V) > 1 + \frac{2\mathbb{E}[U]}{\mathbb{E}[V]}, \quad (6)$$

which reduces to the familiar  $\text{CV}^2(V) > 1$  only in the sparse-duty limit  $\delta \rightarrow 0$  ( $\mathbb{E}[U]/\mathbb{E}[V] \rightarrow 0$ ); for the plotted parameters ( $\mathbb{E}[U]/\mathbb{E}[V] = 1/9$ ) the threshold is  $\text{CV}^2 \approx 1.22$ , i.e.  $\text{CV} \approx 1.11$ . Peak age can therefore sit below average age on heavy-tailed terrestrial links (Fig. 10)—a single freshness number is misleading across heterogeneous segments. The order-statistic replication analysis below acts on the residual law abstractly; the energy-threshold structure extends when the resulting gain profile and firing process satisfy the conditions of Section 6.

#### 5. Result 2: Phase Mixing and Two-Copy PAoI

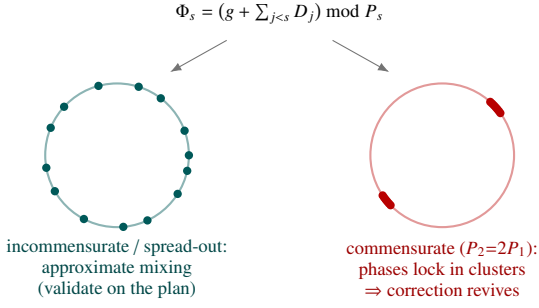
Composing segments needs care. The residual wait  $R_s$  a bundle incurs at segment  $s$  depends on *when* it reaches that segment relative to the segment's own schedule—its arrival phase—which is itself displaced by all the upstream delays. Writing the phase as  $\Phi_s = (g + \sum_{j<s} D_j) \bmod P_s$  (generation time  $g$  plus the accumulated increments, reduced modulo the segment period  $P_s$ ),  $R_s$  is therefore not automatically independent of the upstream increments  $\{D_j\}_{j<s}$ , and the per-segment delays cannot simply be added. The following lemma gives the condition under which they can.

**Lemma 1** (Uniform phases under random offsets). *Let each segment's schedule carry an offset  $\Theta_s \sim \text{Unif}[0, P_s]$ , independent across segments and of the generation process (the random-offset ensemble). Then the phases  $\{\Phi_s\}$  are exactly uniform and mutually independent; hence  $\{D_s\}$  are independent, the aged-update correction vanishes, and  $f_Y = f_{D_1} * \dots * f_{D_S}$ .*

*Proof.*  $\Phi_s = (g + \sum_{j<s} D_j - \Theta_s) \bmod P_s$ ; conditioning on the generation time and all upstream increments, the wrap of an independent uniform  $\Theta_s$  is uniform on  $[0, P_s)$  and independent of the conditioning variables and of the other  $\Theta_j$ .  $\square$

**Remark 3** (Fixed plans: two asymptotic mechanisms, one approximation). *For one fixed plan (deterministic offsets), uniformity and independence are not automatic; they arise in two limits whose limiting parameter we now name explicitly. (a) Horizon limit, deterministic Weyl. Under paced generation at spacing GEN with GEN/ $P_1$  irrational, the first-segment phase sequence  $n \text{ GEN} \bmod P_1$  equidistributes over the first  $n$  updates as  $n \rightarrow \infty$  (Weyl)—this part is rigorous, and it is what justifies the uniform-arrival marginal on the entry segment of a fixed plan. We stress its limit: a downstream phase is not linear in  $n$ —it is the generation epoch pushed through the composed store-carry-forward map, whose increments are themselves determined by the upstream schedules—and joint equidistribution of the generation phases does not by itself imply statistical independence of the downstream residuals. We do not prove measure-preservation for the composed contact map; downstream decoupling on fixed plans rests on mechanism (b) or on measurement. (b) Convolution limit, spread-out delays. If the upstream sum  $\sum_{j<s} D_j$  of  $m$  random increments has a spread-out (non-lattice) component, its characteristic function satisfies  $|\mathbb{E} e^{2\pi i n \Sigma / P_s}| \rightarrow 0$  as  $m \rightarrow \infty$  for all  $n \neq 0$ , so the wrapped phase of a single update converges to uniform with hop depth—this gives the uniform marginal at segment  $s$ ; mutual independence across segments is again the measured part. Large dispersion alone is not sufficient: a lattice-valued sum concentrated on multiples of  $P_s$  has arbitrarily large variance yet a degenerate wrapped phase. At any finite horizon and hop count the fixed-plan use of the uniform law is therefore an engineering approximation, whose accuracy we measure rather than assume: KS-to-uniform 0.001 in the fixed-plan simulations, per-seed composition error  $\leq 5.9\%$  on the heterogeneous chain, and—as the cautionary bound—real Iridium pairs whose residual correlations range from  $-0.51$  to  $+0.88$  when condition (a) is violated by near-commensurate passes (Section 9). The named failure modes are (a') commensurate periods (a UAV synchronized to one specific LEO pass) and (b') lattice-concentrated upstream delays (Fig. 4); Lemma 1 (random offsets) remains the exact statement.*





**Fig. 4.** Phase mixing (Lemma 1, Remark 3). The arrival phase at segment  $s$  lives on a circle of circumference  $P_s$ . Under random schedule offsets the wrapped phase is exactly uniform and independent of the upstream increments; for a fixed plan with spread-out (non-lattice) upstream delays and incommensurate periods the same decoupling holds *approximately* (left)—the regime in which latencies compose by convolution. Commensurate periods lock the phase into clusters (right): the failure mode of a UAV synchronized to one specific LEO pass.

Under paced generation with order-preserving deliveries (Section 3;  $Y_{\max} - Y_{\min} \leq Z_{\min}$  suffices),  $\text{PAoI}^{\text{per}} = Z + Y$  and the single-copy mean needs no independence:

$$\mathbb{E}[\text{PAoI}^{\text{per}(1)}] = \mathbb{E}[Z] + \sum_{s=1}^S (\mathbb{E}[R_s] + T_s), \quad (7)$$

with deterministic envelope  $\text{PAoI}^{\text{per}(1)} \leq Z_{\max} + \sum_s (G_s + T_s)$ .

**Theorem 2** ( $k=2$  per-update PAoI, phase-mixing regime). *With i.i.d. copy latencies  $Y^{(1)}, Y^{(2)} \sim F_Y$  and  $Y_{\min} = \min(Y^{(1)}, Y^{(2)})$ ,*

$$\mathbb{E}[\text{PAoI}^{\text{per}(2)}] = \mathbb{E}[Z] + \int_0^\infty [1 - F_Y(y)]^2 dy, \quad (8)$$

$$\mathbb{E}[Y_{\min}] = \mathbb{E}[Y] - \frac{1}{2} \mathbb{E}[Y^{(1)} - Y^{(2)}] \leq \mathbb{E}[Y], \quad (9)$$

with equality iff  $Y$  is deterministic.

**Remark 4** (Cadence, dispersion, and what replication buys). *Replication leaves the generation cadence unchanged (one reset per delivered update, at its earliest surviving copy; an update losing all copies produces none), so in the lossless limit the mean inter-delivery interval is invariant,  $\mathbb{E}[Z'] = \mathbb{E}[Z]$ , and replication acts only through  $Y_{\min}$ —whose saving is exactly half the mean absolute difference of path latencies (the standard Gini-mean-difference identity): largest at sparse, high-dispersion contacts, nearly nil on near-deterministic links. The invariance is a mean-rate statement only: with losses or reordering, replication also tightens the distribution of  $Z$  (fewer long fault-induced outage gaps)—indeed the Tier-2 gain (Section 9) comes largely from eliminating such gaps, not from  $Y_{\min}$  alone. Shrinking the mean cadence itself requires  $k$  independent update streams, a different energy trade.*

**Remark 5** (Worked ratio and general  $k$ ). *For two ensemble-phase satellites with  $Y \approx R + T_s$ ,  $\mathbb{E}[R_{\min}]/\mathbb{E}[R] = \frac{2}{3}(1 - \delta)$ —a  $\approx 33\%$  cut of the gating term at sparse contacts (Fig. 7).  $\mathbb{E}[Y_{\min,k}] = \int_0^\infty (1 - F_Y)^k$  generalizes, but closed forms past  $k=2$  with multi-segment convolutions are unwieldy; we use a recursive numerical evaluation for general  $k$ . The minimum identity assumes copies traverse independent contacts; same-bottleneck copies give  $Y^{(1)} = Y^{(2)}$  and zero gain.*

## 6. Result 3: Energy–Replication Threshold

Replicating a bundle over several contacts lowers its delivered latency (the freshest copy wins) but each copy drains the relay’s battery, so there is a best number of copies to send—the question this section answers. Deterministic, common service  $T_s$  is an additive constant in the peak, so the only  $k$ -dependent term is the LEO gating term. The

delivered residual over  $k$  contacts is, in full generality, an integral of the *joint* survival,

$$\mathbb{E}[R_{\min,k}] = \int_0^\infty \Pr(R_1 > r, \dots, R_k > r) dr, \quad (10)$$

and how it evaluates depends on the plan model—the  $k$  residuals are functions of a *common* arrival instant, so they are not automatically independent even for staggered windows:

- (i) *Random-offset ensemble.* If each copy’s schedule offset  $\Theta_i$  is an independent uniform on  $[0, P)$ , the wrapped phases  $U - \Theta_i$  are i.i.d. uniform, the  $R_i$  are *exactly* i.i.d. with the residual law of Section 4, and (10) factorizes:

$$\mathbb{E}[R_{\min,k}] = \int_0^\infty \prod_{i=1}^k \Pr(R_i > r) dr \quad (11)$$

$$= \int_0^G \left(\frac{G-r}{P}\right)^k dr = \frac{P(1-\delta)^{k+1}}{k+1}, \quad (12)$$

with  $G = P(1-\delta)$ . (Consistency:  $k=1$  recovers Theorem 1; heterogeneous marginals use (11) directly.) For a *fixed* plan whose copy routes have pairwise incommensurate periods, the relative phases drift, and the same product form is a *time-average approximation* over the horizon—not exact independence, since all residuals share the arrival instant (Remark 3); its accuracy is checked on the plan, and otherwise the exact joint form (10) applies.

- (ii) *Fixed co-period plan.* With deterministic offsets the joint survival must be evaluated on the plan—it is deterministic and *computable at planning time*. For  $k$  windows evenly staggered by  $P/k$  the union schedule is a single comb of period  $P/k$ , so (with  $x_+ = \max\{x, 0\}$ )

$$\mathbb{E}[R_{\min,k}] = \frac{P}{2k} (1 - k\delta)_+^2, \quad (13)$$

*better* than the ensemble value (12): deterministic anti-phasing beats blind placement. Once  $k\delta \geq 1$  the staggered comb covers the entire period: the residual is zero, the gain profile is flat, and further copies cannot improve the gating term—the smallest covering degree  $\lceil 1/\delta \rceil$  is the energy-conserving choice; the threshold analysis below operates on the strictly decreasing range  $k\delta < 1$ .

- (iii) *Locked phases.* Coincident offsets give  $\mathbb{E}[R_{\min,k}] = \mathbb{E}[R]$ : zero gain.

The extremal fixed plans and the ensemble form a reference ladder—the *phase ladder*

$$\underbrace{\frac{P}{2k}(1 - k\delta)^2}_{\text{even stagger}} \leq \underbrace{\frac{P(1-\delta)^{k+1}}{k+1}}_{\text{phase-agnostic}} \leq \underbrace{\frac{P}{2}(1 - \delta)^2}_{\text{locked}}, \quad (14)$$

with equality throughout at  $k=1$ . The outer terms are the extremal fixed plans (best and worst deterministic offsets); the middle term is the *ensemble* reference, not a separator—an individual fixed plan lands anywhere in the outer interval, on either side of it. This is exactly the ordering measured on real Iridium-NEXT pairs in Section 9 (residual-min ratio 0.41 anti-phased, 0.61 ensemble reference, 0.95 near-locked, with intermediate pairs scattered on both sides). We develop the threshold below for the *phase-agnostic* profile (i)—the operative model when copy routes have drifting relative phases or are assigned without phase information—and note that every structural result carries over to profile (ii) with (13) in place of (12) on its strictly decreasing range  $k\delta < 1$  (beyond full coverage the profile is flat and the smallest covering degree is optimal).

*Energy-adequacy ratio.* Measuring energy in copies (harvest  $\rho = \lambda_e/e$  copies per unit time, contact rate  $\lambda_c = 1/P$ ), define

$$\eta \triangleq \frac{\rho}{\lambda_c} = \frac{\lambda_e P}{e} = \frac{\lambda_e/\lambda_c}{e} \quad (\text{copies harvested per period}). \quad (15)$$

Demanding  $k$  copies per period caps the launchable fraction of update epochs at  $p_e(k) = \min(1, \eta/k)$  (flow balance; derived exactly in Section 7). Under paced generation with all-or-nothing firing the mean per-update peak is  $\mathbb{E}[\text{PAoI}^{\text{per}}] = P/p_e + \mathbb{E}[R_{\min,k}] + T_s$  (a firing every  $1/p_e$  periods on average, plus the delivered residual); dropping the  $k$ -independent constant  $P + T_s$  gives the mean-PAoI<sup>per</sup> gating objective

$$\mathcal{A}(k) = P \underbrace{\frac{1 - p_e(k)}{p_e(k)}}_{\text{starvation}} + \underbrace{\frac{P(1 - \delta)^{k+1}}{k+1}}_{\text{gain}}, \quad p_e(k) = \min\left(1, \frac{\eta}{k}\right). \quad (16)$$

For  $k \leq \eta$  the starvation term vanishes and  $\mathcal{A}$  is strictly decreasing; for  $k > \eta$  the starvation term equals  $\frac{P}{\eta}(k - \eta)$  and rises linearly while the gain term decays. Hence  $\mathcal{A}$  is unimodal and its minimizer is one of the two integers bracketing  $\eta$ :

$$k^* = \min(K_{\max}, k^\circ), \quad k^\circ \in \{\lfloor \eta \rfloor, \lceil \eta \rceil\}, \quad \eta = \frac{\lambda_e P}{e}. \quad (17)$$

Here  $\lfloor \eta \rfloor$  is the largest degree that incurs *no* starvation—the conservative DTN-reliability rule—while the skip-aware optimum takes the extra copy,  $k^\circ = \lceil \eta \rceil$ , iff its added starvation is repaid by the order-statistic gain:

$$P\left(\frac{\lceil \eta \rceil}{\eta} - 1\right) < g_{\lfloor \eta \rfloor} - g_{\lceil \eta \rceil}, \quad g_k = \frac{P(1 - \delta)^{k+1}}{k+1}. \quad (18)$$

(For integer  $\eta$ ,  $k^* = \eta$ ;  $k^* = 1$  if  $\eta < 1$ .)  $K_{\max}$  is the number of available copy routes. The distinction between the conservative rule  $\lfloor \eta \rfloor$  and the skip-aware optimizer is itself a contribution: occasional energy-forced skips can be worth an extra copy near the top of an integer interval.

**Remark 6** (Conditionality and gain-profile generality). *Everything above is conditional on (a) all-or-nothing firing and (b) the phase-agnostic gain profile  $g_k = P(1 - \delta)^{k+1}/(k+1)$  of plan model (i). The structure generalizes with one quantitative caveat. For any strictly decreasing gain profile with diminishing increments ( $g_k - g_{k+1} \geq g_{k+1} - g_{k+2}$  for all  $k$ ),  $\mathcal{A}$  is unimodal (the starvation term is zero up to  $\eta$  and rises linearly beyond it), and Theorem 3's monotonicity holds. The two-candidate bracketing  $k^\circ \in \{\lfloor \eta \rfloor, \lceil \eta \rceil\}$  additionally requires that the first forward gain drop beyond the threshold not exceed the starvation slope,*

$$g_{\lfloor \eta \rfloor} - g_{\lceil \eta \rceil} \leq \frac{P}{\eta}; \quad (19)$$

*diminishing increments then keep  $\Delta_k \mathcal{A} \geq 0$  for all  $k \geq \lceil \eta \rceil$ . Without (19) the rule can fail (e.g.  $g_k \propto c/k$  with large  $c$  places the optimum far above  $\lceil \eta \rceil$  while satisfying diminishing increments). Both plan models satisfy (19) automatically: for (i),  $g_k \leq P/(k+1)$ , and for the even-stagger profile  $g_k^{\text{stag}} = \frac{P}{2k}(1 - k\delta)^2$  of plan model (ii) (feasible range  $k\delta \leq 1$ ),  $g_k^{\text{stag}} \leq P/(2k)$ ; in both cases the drop at  $k = \lceil \eta \rceil$  is below  $P/\lceil \eta \rceil \leq P/\eta$  since  $\eta \leq \lceil \eta \rceil$ . The same floor/ceiling policy therefore applies to a deterministically staggered plan with  $g^{\text{stag}}$  replacing  $g$  in (18)—only the crossover points move. For a general fixed plan,  $g_k$  is computed from (10) at planning time and (19) is checked alongside.*

**Theorem 3** (Monotonicity and saturation). *For any strictly decreasing gain profile  $g_k$  with diminishing increments,  $k^*(\eta)$  is non-decreasing in  $\eta = \lambda_e/(\lambda_e e)$ , and is constant at  $K_{\max}$  for all  $\eta \geq K_{\max}$ .*

*Proof.* Write  $\mathcal{A} = g + h$  with  $h(k; \eta) = P \max(0, k/\eta - 1)$ . The forward difference  $\Delta_k g < 0$  is independent of  $\eta$ , while  $\Delta_k h(k; \eta) = \frac{P}{\eta} \min(1, (k+1 - \eta)^+) \geq 0$  (equal to  $P/\eta$  for  $k \geq \eta$ , to 0 for  $k+1 \leq \eta$ , linear between) is non-increasing in  $\eta$  for each fixed  $k$ . Hence  $\Delta_k \mathcal{A}$  is non-increasing in  $\eta$ , so the first up-crossing  $k^*(\eta) = \min(\{k : \Delta_k \mathcal{A} \geq 0\} \cup \{K_{\max}\})$  is non-decreasing in  $\eta$ . For  $\eta \geq K_{\max}$ ,  $\Delta_k h = 0$  over the feasible range, so  $\mathcal{A} = g$  is strictly decreasing and the minimizer pins to  $K_{\max}$ .  $\square$

**Remark 7** (Interpretation).  $k^*$  tracks the harvest-to-contact ratio: launch as many independent copies as the energy budget sustains without forcing skipped windows, and no more. Over-replication ( $k > \eta$ )

*drains the battery and starves future contacts, so the linear starvation term overwhelms the vanishing order-statistic gain. When harvest covers full contact diversity ( $\eta \geq K_{\max}$ ) energy stops binding and  $k^* = K_{\max}$ ; when  $\eta < 1$ ,  $k^* = 1$  (no replication), matching the energy-harvesting AoI result that aggressive replication is sub-optimal once harvest rate falls below the update rate.*

**Remark 8** (Where the replication gain comes from). *Deterministic scheduling by itself produces no strict replication benefit: under a fault-free, exactly known contact plan, an oracle single-copy router (CGR with the true plan) already selects the earliest-arriving route, so  $\min_i Y^{(i)}$  is attained with one copy and higher degrees can only tie—our Tier-2 experiment confirms this on a fault-free staggered diamond (Section 9). The strict gain characterized in this section is operative precisely under route-outcome uncertainty, of which our framework instantiates three forms: (a) plan uncertainty—the phase-agnostic (random-offset) model treats the copy phases as unknown at decision time, so no single route is knowably earliest; (b) outcome uncertainty—relay faults break the nominally earliest route (Tier-2's 6 $\times$  gain is of this type); and (c) prediction error in the contact plan itself (Section 8). The deterministic residual and renewal laws are thus the structure against which the gain is computed—they set the gain profile  $g_k$ —while uncertainty is the source of the gain. This is also why predictable space-DTN routing conventionally avoids duplication: with no uncertainty, it buys nothing.*

*Uncertainty-aware threshold.* Remark 8 identifies uncertainty as the source of the gain; we now fold it into the optimization itself, since (16) is a *lossless* analysis in which every fired update eventually resets the age. Let each copy *succeed* with probability  $q$  (its route materializes and delivers), with energy spent on the launch regardless; successes are i.i.d. across copies and periods and independent of the copies' contact phases and residuals. An admitted update then succeeds with probability  $s_k = 1 - (1 - q)^k$ , and the delivered residual is the minimum over the  $\text{Bin}(k, q)$  surviving copies:

$$h_k(q) = \frac{P[(1 - q\delta)^{k+1} - (1 - q)^{k+1}]}{q(k+1)} - G(1 - q)^k, \quad (20)$$

the conditional mean of  $R_{\min}$  given at least one survivor (phase-agnostic profile;  $h_k(1) = g_k$  and  $h_1(q) = \mathbb{E}[R]$ ).

**Proposition 1** (Peak age under route uncertainty). *Under paced generation, all-or-nothing firing, order-preserving successful deliveries (Section 3), and the i.i.d. success model above: (i) in stationarity the successful-update rate is  $p_e s_k$  per period, so*

$$\begin{aligned} \mathbb{E}[\text{PAoI}^{\text{per}}] &= \frac{P}{p_e(k) s_k} + h_k(q) + T_s, \\ \mathcal{A}_q(k) &= P \frac{1 - p_e s_k}{p_e s_k} + h_k(q), \end{aligned} \quad (21)$$

*by the stationary successful-update rate (the inter-success spacing is geometric only in the i.i.d.-admission special case; the mean needs only the rate); (ii)  $\mathcal{A}_q$  is strictly decreasing on  $k \leq \lfloor \eta \rfloor$  (there  $p_e = 1$ ,  $s_k$  is strictly increasing, and  $h_k$  is strictly decreasing); and (iii) for  $k \geq \eta$  the cadence increment is nonnegative,*

$$\Delta_k \left( \frac{k}{s_k} \right) = \frac{s_k - kq(1 - q)^k}{s_k s_{k+1}} \geq 0, \quad (22)$$

*since  $s_k = q \sum_{j < k} (1 - q)^j \geq kq(1 - q)^{k-1} \geq kq(1 - q)^k$ ; hence the minimizer lies in  $\{\lfloor \eta \rfloor, \lceil \eta \rceil\}$  whenever the increment condition*

$$\frac{P}{\eta} \Delta_k \left( \frac{k}{s_k} \right) \geq h_k(q) - h_{k+1}(q) \quad \forall k \geq \lceil \eta \rceil \quad (23)$$

*holds—the uncertainty analog of (19). At  $q = 1$ , (23) reduces to (19) and the proposition recovers (16)–(17) exactly.*

Throughout, the candidate set is clipped to  $[1, K_{\max}]$  as in (26): for  $\eta < 1$  the floor is infeasible and  $k=1$  is the lower candidate. Parts (i)–(iii) are proved; what is *not* claimed as a theorem is that (23) holds universally. Since (23) depends on  $\delta$  through  $h_k(q)$ , we verified it—and the resulting clipped two-candidate optimum—over the full grid  $\delta \in [0.02, 0.9] \times q \in [0.05, 1] \times \eta \in [0.1, 8]$  (840 points,  $k \leq 12$ ): the condition holds with minimum margin  $0.067P$  (attained at small  $\delta$ , small  $q$ , large  $\eta$ ) and the optimizer never left the candidate set (check\_gainboundq.py in the artifact); E18 (Section 9) confirms the same in simulation. For parameters outside these ranges (23) is a one-line numerical check performed alongside the policy, exactly as (19) is for fixed plans.

The structural change at  $q < 1$  is that replication now buys *cadence*, not only latency spread: the failure-gap term  $P(1 - s_k)/s_k$  decays geometrically in  $k$ . Here  $q$  is the *effective per-pass route-success probability*—emphatically not the relay’s duty-cycle availability: the unified Tier-2 experiment (Section 9) shows a relay that is down 1/3 of the time yet delivers 100% with bounded delay inflation—consistent with an effective  $q$  close to one (eventual delivery alone does not pin the per-pass value; a copy can miss its pass, re-route, and still arrive)—while the route-poor two-relay setting of Fig. 13 converts mid-pass stalls into missed passes and realizes the large gain. In practice  $\hat{q}$  is estimated directly, as the fraction of admitted single-copy updates that succeed on their operative pass; the observed inter-success cadence then cross-checks it via  $\mathbb{E}[Z_{\text{succ}}] = P/(p_e \hat{q})$ . We do not quote a numerical  $\hat{q}$  from the OPAoI trace—OPAoI is a cycle maximum, not a per-update spacing—but directionally, the route-poor experiment’s mean OPAoI  $\approx 7P$  against a 2/3 relay uptime puts the effective per-pass  $q$  far below availability, quantifying how route poverty converts stalls into losses. The i.i.d. Bernoulli model is then an illustrative first-order bridge, and fault-dwell correlation (failures clustering across consecutive passes) is the known refinement it omits. The contact-prediction experiment of Section 8, by contrast, *is* exactly the i.i.d.  $q$ -model with  $q = \alpha$  by construction: evaluating (21) at its parameters reproduces the observed ordering (keeping  $k=\lfloor \eta \rfloor$  beats the reserve heuristic  $\lfloor \alpha \eta \rfloor$ , e.g.  $\mathcal{A}_{0.4}(4) = 0.45P$  versus  $\mathcal{A}_{0.4}(2) = 0.93P$  at  $\eta=6$ )—so that empirical finding is a prediction of the theory.

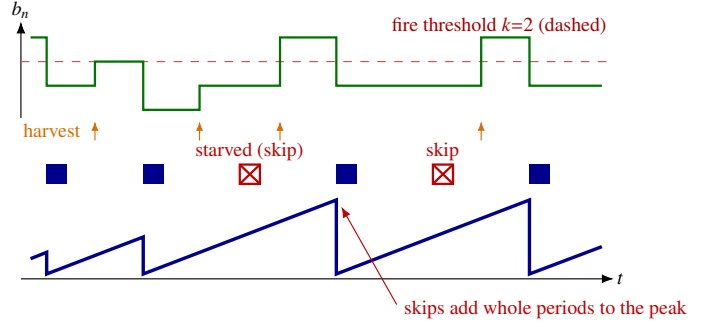
## 7. Hardening: Exact Battery-Queue Throttle

The throttle  $p_e = \min(1, \eta/k)$  used above was a fluid approximation. In reality the relay’s battery is a finite queue that fills as energy is harvested and drains  $k$  units each time it fires  $k$  copies (Fig. 5); we now replace the fluid throttle by the stationary law of this energy queue and couple it to the age dynamics.

**Energy queue.** Let the battery hold up to  $B$  units (one unit = the energy  $e$  to send one copy) and harvest  $A$  units per contact period, i.i.d. across periods with mean  $\mathbb{E}[A] = \eta$  (the harvest-per-period in copies; Poisson for the explicit chain). Under the all-or-nothing policy—fire  $k$  copies iff  $b \geq k$ , else hold and recharge—the embedded chain at contact epochs is

$$b_{n+1} = \min(B, (b_n - k \mathbb{1}\{b_n \geq k\}) + A_n), \quad (24)$$

a finite chain on  $\{0, \dots, B\}$ . For Poisson harvest (support all of  $\mathbb{N}_0$ ) the chain is irreducible and aperiodic on  $\{0, \dots, B\}$  with a unique stationary  $\pi$ ; the throttle is  $p_e(k) = \mathbb{P}_\pi(b \geq k)$ . No simple positivity condition substitutes for this support assumption:  $\Pr(A \geq 1) > 0$  alone fails ( $A \in \{0, 2\}$  with  $k=2$  traps a parity class), and even  $\Pr(A=0) > 0$  with  $\Pr(A=1) > 0$  fails (Bernoulli  $A \in \{0, 1\}$  with  $k=2$  never accumulates past  $b=k$ , leaving the states above  $k$  unreachable). In general we assume the harvest support induces an irreducible, aperiodic transition graph on the reachable battery states; otherwise—including lattice or deterministic harvest (e.g.  $A \equiv \eta \in \mathbb{N}$  cycles deterministically)— $\pi$  and  $p_e$  are read on the recurrent class reachable from  $b_0$ , which exists by finiteness, and the work-conservation identity below holds under *any* stationary regime, since it is a flow balance that never invokes ergodic structure.



**Fig. 5.** Energy-gated firing, sample path ( $k=2$ ). *Top:* the battery fills by harvest and drops  $k$  units at each firing; a window that opens with  $b < k$  is *starved* and skipped (crossed). *Bottom:* the age resets only at fired windows, so each skip adds a full period to the peak  $JP + R_{\min} + T_s$ —the starvation term  $P(1 - p_e)/p_e$  of (16). After a firing the battery is depleted, which is why consecutive windows are correlated (Markov-renewal gaps) and the fluid throttle is a mean-field limit.

**Lemma 2** (Work conservation; distribution-free). *Each firing slot consumes exactly  $k$  units, so in steady state  $k p_e(k) = \eta - L$  with  $L \geq 0$  the mean per-period harvest lost to the cap. Hence*

$$p_e(k) = \frac{\eta - L}{k} \leq \min\left(1, \frac{\eta}{k}\right), \quad (25)$$

with  $L = 0$  iff no harvested energy is truncated by the cap (the battery may reach  $B$  exactly without loss; only overflowed harvest counts).

Only the mean  $\eta$  enters at this order; the harvest distribution and  $B$  affect  $p_e$  solely through  $L$ . Thus the throttle of Section 6 is exactly the no-overflow / large- $B$  limit, and the finite-battery correction is sign-definite ( $p_e$  can only be smaller).

**Corollary 1** ( $k = 1$  closed form). *Let the per-period harvest increments  $A_n$  be i.i.d., nonnegative, with  $\mathbb{E}[A_n] = \eta$  and  $\mathbb{P}(A_n=0) > 0$ , so the chain  $b_{n+1} = (b_n - 1)^+ + A_n$ —the embedded queue-length chain of a discrete-time deterministic-unit-service queue—is ergodic on the recurrent class reachable from  $b_0$  for  $\eta < 1$  (all of  $\mathbb{N}_0$  under unbounded support, e.g. Poisson; trivially for finite  $B$ ); the stated marginals below follow from flow balance and are class-independent. For  $B \rightarrow \infty$  and  $\eta < 1$ ,  $\mathbb{P}_\pi(b = 0) = 1 - \eta$ , so  $p_e(1) = \eta$ , with the Pollaczek–Khinchine level law. Finite  $B$  gives  $L \geq 0$  and  $p_e(1) \leq \eta$ , with strict loss exactly when overflow has positive stationary probability—e.g. under Poisson harvest for any finite  $B$ —but not universally: for Bernoulli harvest  $A_n \in \{0, 1\}$  and  $B \geq 1$  the battery can never overflow, so  $L = 0$  and  $p_e(1) = \eta$  exactly despite the finite cap.*

**Consequence for  $k^*$ .** The objective (16) keeps its form—the starvation term  $P(1 - p_e)/p_e$  is exact in  $p_e$  by the long-run firing rate—but now with the true  $p_e(k) = (\eta - L(B, k))/k \leq \min(1, \eta/k)$ . Finite battery thus weakly enlarges each starvation term, exerting *downward pressure* on the optimal degree; empirically (E11)  $k^*$  decreases as  $B$  shrinks (at  $\eta=4$ :  $4 \rightarrow 2$  as  $B$  falls  $64 \rightarrow 4$ ). This is *not* a universal bound  $k^* \leq \lfloor \eta \rfloor$ , however: because the overflow loss  $L(B, k)$  depends on  $k$ , the pointwise bound on  $p_e$  does not order  $\mathcal{A}(k)$  across  $k$ , and a degree that strictly wins in the infinite-battery limit (e.g.  $\lfloor \eta \rfloor$  via (18)) still wins for all sufficiently large  $B$  (as  $B \rightarrow \infty$ ,  $L \rightarrow 0$ ). The finite-battery effect is therefore a quantified downward shift, not a hard ceiling at  $\lfloor \eta \rfloor$ . In our i.i.d.-harvest scenarios, finite  $B$  also raises the variance of skip gaps (bursty starvation: a firing depletes the battery, so skips cluster), inflating the PAoI tail but not the mean; we do not claim this direction universally—under correlated harvest or other admission policies the gap variance can move either way, and only the work-conservation identity for the firing fraction is distribution-free.

**Regime-ODE coupling.** Stationary battery–age coupling (steady-state averaging; no timescale-separation limit is claimed): the battery chain is taken at its stationary  $\pi$  while age evolves under the gated renewal process. The SHS mode is  $q = (\phi, m)$  with  $\phi$  the contact ON/OFF phase and



$m$  the battery regime; the age ODE  $\dot{\Delta} = 1$  has delivery resets enabled only when  $m \geq k$ , i.e.  $I^{\text{eff}}(t) = I(t) \mathbb{1}\{b(t) \geq k\}$ . Averaging the gate over the stationary  $\pi$  yields the thinned alternating-renewal schedule of (4): usable windows recur after  $V' = (J-1)P + G$  with  $\Pr(J=j)$  the stationary firing-gap law. Under i.i.d. firing  $J$  is geometric and every Result-1 form extends exactly through the renewal law with  $V \rightarrow V'$ ; with a finite battery the firing gaps are Markov-renewal rather than geometric—after a firing the battery is depleted, so usable windows cluster—and the geometric- $J$  forms become a mean-field approximation. Work conservation (Lemma 2) still pins the firing fraction at  $p_e$  exactly, so the mean starvation term is exact in  $p_e$ ; what the correlation moves is the firing-gap variance (hence the peak-age tail), which requires the correlated chain. A minimal two-regime reduction ( $m \in \{b < k, b \geq k\}$ ,  $\mathbb{P}(\text{hi}) = p_e$ ) recovers the mean; the full  $(B+1)$ -mode generator gives the correlated refinement.

## 8. An AoI-Energy Replication Policy

The threshold of Section 6 yields a concrete policy that layers on top of an existing contact-plan router (CGR/RUCoP). CGR computes *which* routes a bundle can take; the policy decides *how many* copies to commit, given its long-run energy adequacy. A source that knows its harvest rate  $\lambda_e$ , per-copy cost  $c$ , contact cadence (hence  $\eta = \lambda_e P/e$ ), and the number  $K_{\max}$  of candidate copy routes CGR offers selects the replication degree by a closed-form *two-candidate* rule. By (17)–(18) the unconstrained optimum is one of the two integers bracketing  $\eta$ , so the policy evaluates  $\mathcal{A}$  at just those two and clips to  $[1, K_{\max}]$ :

$$k_{\text{pol}}(\eta) = \arg \min_{k \in \mathcal{K}(\eta)} \mathcal{A}(k), \quad (26)$$

$$\mathcal{K}(\eta) = \{\text{clip}(\lfloor \eta \rfloor, 1, K_{\max}), \text{clip}(\lceil \eta \rceil, 1, K_{\max})\},$$

with  $\mathcal{A}$  from (16) (or  $\mathcal{A}_q$  from (21) when a route-success estimate is available); it then commits the  $k_{\text{pol}}$  copies and spends only  $k_{\text{pol}} e$  per update. The choice is local and  $O(1)$  (two objective evaluations, no sweep over  $k$ ). It is driven by the *mean* energy-adequacy  $\eta$ —a slow estimate of harvest rate and cadence—rather than the instantaneous battery level, and is recomputed only when that estimate drifts; it is thus a steady-state set-point, not a per-window controller.

*Joint degree and placement.* Which  $k_{\text{pol}}$  routes matters as much as how many: “distinct next-hops” do not make copy latencies independent—routes can share source contacts, relays, downstream links, failure domains, or (per the Iridium scan of Section 9) near-commensurate orbital phases. Moreover, since replication pays only under route uncertainty, the placement objective must weigh *timing* diversity and *failure* diversity together. Both live in one quantity: for a candidate route set  $\mathcal{S}$  with per-route failure events  $F_i$  (success probability  $q_i$ , possibly correlated across routes) and plan-determined residuals  $R_i$ , the set’s success probability is  $s_{\mathcal{S}} = 1 - \Pr(\bigcap_{i \in \mathcal{S}} F_i)$  and the delivered residual given success generalizes (10) to

$$\mathbb{E}[R_{\min, \mathcal{S}}^{\text{succ}}] = \frac{1}{s_{\mathcal{S}}} \int_0^{\infty} \left[ \Pr\left(\bigcap_{i \in \mathcal{S}} (F_i \cup \{R_i > r\})\right) - \Pr\left(\bigcap_{i \in \mathcal{S}} F_i\right) \right] dr. \quad (27)$$

Under the common periodic firing epoch, single-energy-bottleneck, constant per-copy-cost model, the exact planning-time problem is the joint combinatorial one,

$$\min_{\emptyset \neq \mathcal{S} \subseteq \mathcal{R}, |\mathcal{S}| \leq K_{\max}} \left[ \frac{P}{p_e(|\mathcal{S}|) s_{\mathcal{S}}} + \mathbb{E}[R_{\min, \mathcal{S}}^{\text{succ}}] \right], \quad (28)$$

in which the best set can change the best *degree* (a pair with poor common-mode reliability can lose to a three-route set with better failure-domain diversity); for heterogeneous routes—distinct periods, route-dependent energy, route-specific launch epochs—the cadence

term  $P/(p_e s_{\mathcal{S}})$  is replaced by the set’s actual mean inter-successful-update interval. We do not solve (28) here. The precise reading of our components is: (26) is the  $O(1)$  degree set-point that is exact for a statistically *exchangeable* route pool (i.i.d. successes, ensemble phases, where (28) reduces to (21)); the (27)-based set choice at fixed  $|\mathcal{S}|$  is a placement *refinement*; and fully joint heterogeneous degree-and-placement optimization is an extension. Note the  $O(1)$  claim covers the degree rule only—evaluating (27) over route subsets is combinatorial, with disjointness and pairwise-correlation proxies as the cheap surrogates. The two diversities can trade off: an anti-phased pair sharing a failure domain (correlated  $F_i$ , small  $s_{\mathcal{S}}$ ) can lose to a slightly less anti-phased pair with independent failures—timing diversity optimizes  $\mathbb{E}[R_{\min}^{\text{succ}}]$ , failure diversity optimizes  $s_{\mathcal{S}}$ , and (27) prices both. On the real constellation the timing axis alone already spans more than the degree step: the best-aligned Iridium pair reaches a residual-min ratio of 0.41 where a blind pick averages  $\sim 0.65$  and a phase-locked pair returns 0.95.

We evaluate the policy in the finite-battery model against every fixed degree  $k = 1, \dots, K_{\max}$  and against the per- $\eta$  exhaustive optimum (Fig. 11). The closed-form rule (26) matched the exhaustive optimum at all sampled energy levels, and its PAoI is the lower envelope of the fixed-degree curves: any fixed  $k$  is wasteful when energy is scarce (a large  $k$  starves) or stale when it is abundant (a small  $k$  forgoes the order-statistic gain), whereas the adaptive policy is best at every energy level. We stress the scope of this statement: the two-candidate rule is derived in the large-battery model, and there is *no theorem* that the finite-battery optimum stays in  $\{\lfloor \eta \rfloor, \lceil \eta \rceil\}$  in general—it did so in every tested  $(\eta, B)$  configuration, but our own E15 experiment exhibits the boundary: under strongly persistent (dwell $\approx 50$ ) Markov-modulated harvest the optimum moves to  $k=2 < \lfloor \eta \rfloor$  at  $\eta=3$ , outside the candidate set. The policy is thus an empirically robust set-point with a measured failure regime, not a finite-battery optimality theorem. It is realized in a DTN-realistic stack by our CGR-native  $k$ -copy router (Section 9).

*Worked example (Iridium-class numbers).* Take the paper’s LEO scenario:  $P = 5736$  s,  $W = 430$  s ( $\delta = 0.075$ ),  $T_s = 10$  s, and a solar harvest that accumulates one copy’s energy every 2200 s, so  $\eta = \lambda_e P/e = 2.6$ . The two candidates are  $k \in \{2, 3\}$ ; the ceiling test (18) compares added starvation  $P/(3/2.6 - 1) = 882$  s against order-statistic gain  $g_2 - g_3 = 463$  s, so the floor wins:  $k_{\text{pol}} = 2$ . The resulting mean PAoI<sup>per</sup> is  $P + \mathbb{E}[R_{\min, 2}] + T_s = 5736 + 1513 + 10 \approx 7.3 \times 10^3$  s ( $\approx 2.0$  h), versus  $8.2 \times 10^3$  s ( $\approx 2.3$  h) single-copy—a 12% freshness gain for one extra copy—whereas  $k=3$  would *lose* ground ( $\mathcal{A}(3) - \mathcal{A}(2) \approx +419$  s) by starving one window in 7.5.

*Tail / deadline-violation.* For mission-critical traffic the relevant quantity is the PAoI *tail*,  $\Pr(\text{PAoI}^{\text{per}} > \Delta)$ , not the mean—and in the LEO  $k$ -copy model the tail is available in closed form.

**Corollary 2** (PAoI<sup>per</sup> tail, LEO  $k$ -copy). *Under the model of Section 6 with i.i.d. firing (exact when  $p_e = 1$ ; mean-field otherwise), the per-update peak decomposes as  $\text{PAoI}^{\text{per}} = JP + R_{\min, k} + T_s$  with  $J \sim \text{Geom}(p_e)$  independent of  $R_{\min, k}$ . For  $x = T_s + jP + r$ ,  $j \geq 1$ ,  $r \in [0, P)$ ,*

$$\Pr(\text{PAoI}^{\text{per}} > x) = (1 - p_e)^j + p_e(1 - p_e)^{j-1} \left( \frac{G-r}{P} \right)^k \mathbb{1}\{r \leq G\}, \quad (29)$$

*and in the energy-abundant regime ( $p_e=1$ ) the  $\beta$ -quantile is  $T_s + P + \max\{0, G - P(1 - \beta)^{1/k}\}$ ; the maximum handles the atom of  $R_{\min, k}$  at 0 (mass  $1 - (G/P)^k$ , an update generated inside a window), at which the quantile sits for all  $q \leq 1 - (G/P)^k$ .*

Replication thus pulls the tail in polynomially—degree  $k$  in the gap fill—and the simulated CCDF (Fig. 12) matches (29) pointwise: at a  $1.5P$  deadline the closed form gives  $0.405^k = 0.405, 0.164, 0.027$  for  $k=1, 2, 4$  against simulated 0.405, 0.165, 0.027 (a  $15\times$  reduction at  $k=4$ ), and the 99th percentile  $T_s + P + G - P(0.01)^{1/k}$  evaluates to  $1.90P$  ( $k=1$ ) and  $1.59P$  ( $k=4$ ), exactly the simulated quantiles. For a renewal

segment the single-copy tail is immediate from (5):  $\Pr(\text{OPAol} > x) = \Pr(V > x - T_s)$ . The mean alone therefore understates the value of replication; the policy (26) extends to a deadline constraint by minimizing  $\mathcal{A}$  subject to  $\Pr(\text{PAol}^{\text{per}} > \Delta) \leq \epsilon$  with the left side evaluated by (29) rather than simulation.

*Comparison with a RUCoP-style MDP core.* RUCoP/CGR-UCoP casts routing under *uncertain* contact plans as an MDP that maximizes delivery probability for a copy budget. Rather than reproduce the full CGR-UCoP pipeline, we implemented its decision core—a RUCoP-style holder-set backward induction over the time-ordered contacts, budgeting transmissions and validated against the closed form  $1 - (1 - q)^k$  on a symmetric plan—and solved it on a single heterogeneous uncertain diamond to obtain a delivery-vs-copies frontier (an illustrative instance, not a benchmark-scale reproduction). The two approaches are complementary: the MDP answers *which* routes maximize delivery and is energy-blind—its optimum operates at the budget ceiling  $k = K_{\max}$ —whereas our policy answers *how many* copies the energy budget warrants. On this instance the frontier is concave, so copies beyond the knee buy little delivery: at  $k^* = 2$  our policy attains 94.7% of the MDP’s delivery for 40% of the energy (the exact percentages are instance-specific; the concavity, not the number, is the point). The two compose—the MDP’s route set under our degree cap—which is the natural energy-aware extension of CGR-UCoP.

*Robustness to contact-prediction error.* Contact plans are predictions. We model a prediction reliability  $\alpha$ : each committed copy’s predicted contact materializes w.p.  $\alpha$ , with energy spent on the commit regardless (a copy sent to a missed contact is wasted). A natural intuition is to reserve energy under uncertainty—use  $\lfloor \alpha \eta \rfloor$  copies. Our experiments show the opposite: the PAol-optimal degree stays at the no-starvation degree  $\lfloor \eta \rfloor$  for every  $\alpha \in [0.4, 1]$  (here  $\eta$  is set so  $\lfloor \eta \rfloor$  is the model optimum), because extra copies hedge missed contacts through the order statistic over which copies materialize. The rule therefore degrades gracefully (+19% PAol as  $\alpha$  falls to 0.4), whereas the reserve heuristic under-replicates and is counterproductive (+50% PAol at  $\alpha = 0.4$ ). In the tested regime—independent per-copy materialization with energy spent regardless—replication, not a smaller copy budget, is the right response to prediction uncertainty; this is now also a *prediction* of Proposition 1 with  $q = \alpha$  rather than only an observation. With correlated failures, asymmetric route probabilities, or route-dependent energy costs the optimal response can differ, and the  $q$ -model would need the corresponding generalization.

## 9. Numerical Validation

Validation follows a two-tier plan, closed by a reference-implementation cross-check. A model-exact Monte-Carlo simulator confirms the closed forms over the parameter sweeps (duty cycle  $\delta$ , replication degree  $k$ , energy-adequacy  $\eta$ , battery capacity  $B$ ); a DTN-realistic stack with Bundle Protocol and Contact Graph Routing (dtnsim, the contact-plan simulator of the RUCoP/CGR-UCoP lineage), extended with a custom energy-harvesting battery module and a CGR-native  $k$ -copy router, runs the representative scenarios and compares single-copy CGR against CGR replication under an energy budget. A RUCoP-style MDP core is implemented and compared on an illustrative instance in Section 8. Each theoretical claim maps to one experiment with a quantitative pass criterion: the mixed residual law and the quadratic-mean/linear-peak divergence of Theorem 1; the fixed-plan mixing approximation and its phase-locking failure mode (Remark 3); the Gini-dispersion replication gain and its collapse for same-bottleneck copies (Theorem 2); the unimodal objective and threshold  $k^*$  with monotonicity and saturation (Theorem 3); and the work-conservation throttle with the finite-battery correction and its downward pressure on  $k^*$  (Lemma 2). Scenario specifications and contact plans are released for reproducibility.

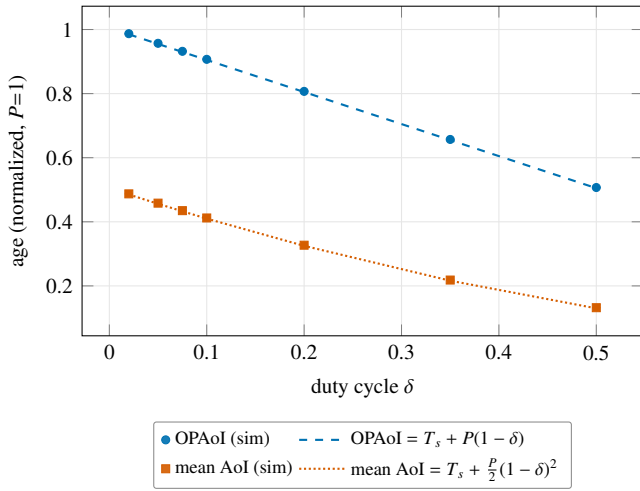
**Table 4**

Tier-1 traceability: analytical claim  $\rightarrow$  model-exact check.

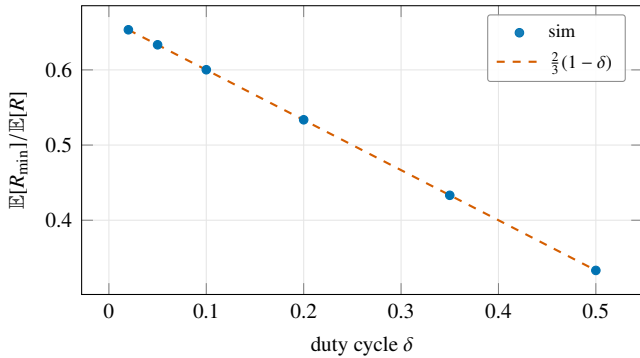
| Claim                     | Check                                                           | Outcome                                                                        |
|---------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------|
| Residual law, Thm 1       | KS to mixed law; mean/peak exponents in $(1-\delta)$            | KS 0.002; slopes 1.98/1.00 (Fig. 6)                                            |
| Fixed-plan mixing, Rem. 3 | phase KS-to-uniform, corr.; commensurate control                | 0.001, $ \rho  = 0.002$ ; locked: KS 0.89                                      |
| Two-copy gain, Thm 2      | Gini identity; ratio $\frac{2}{3}(1-\delta)$ ; same-bottleneck  | matched (Fig. 7); zero gain                                                    |
| Threshold, (17), Thm 3    | unimodality; staircase, saturation                              | argmin brackets $\eta$ (Figs. 8, 9)                                            |
| Ceiling branch (18)       | fine $\eta$ -sweep on $[3, 4]$                                  | flip at $\eta^* = 3.824$                                                       |
| Work cons., Lem. 2        | $k p_e = \eta - L$ ; $k = 1$ P-K law; Poisson vs. deterministic | $\leq 10^{-5}$ ; distribution-free; $p_e(1) = \eta$                            |
| Finite $B$ , Sec. 7       | $k^*$ vs. $B$ (E11)                                             | $4 \rightarrow 2$ as $B$ : $64 \rightarrow 4$ ( $\eta = 4$ )                   |
| Bursty harvest            | Markov-modulated dwell sweep, $\eta = 3$                        | $k^*$ invariant to dwell 10; $w_c \leq 3 \times 10^{-4}$ ; $k = 2$ at dwell 50 |
| Renewal laws (5), (6)     | det., jitter, exp., Pareto; inversion threshold                 | $\mathbb{E}[R]$ within 1%; crossing matched (Fig. 10)                          |
| Uncertainty, Prop. 1      | (21) vs. sim over $q \times k \times \eta$                      | within 0.7%; optimum in candidate set                                          |

*Tier-1 results.* Table 4 traces every analytical claim to its model-exact experiment (time normalized to  $P = 1$ ; 8–20 seeded replications with 95% CIs); all pass. Two rows warrant prose. The *ceiling* branch flips exactly at the analytic crossover of (18) ( $\delta = 0.1$  on  $[3, 4]$ : simulated flip between  $\eta = 3.8$  and  $3.9$ ,  $\eta^* = 3.824$ ), so the extra copy is taken precisely when its gain repays the added starvation. And the bursty-harvest row is the policy’s measured *failure boundary*: the threshold and the work-conservation identity survive Markov-modulated harvest up to a mean dwell of ten periods (correlation moves the tail,  $1.78P \rightarrow 3.07P$  at  $k^*$ —the Markov-renewal effect anticipated in Section 7), but under extreme persistence (dwell  $\approx 50$ ) the optimum retreats to  $k = 2$ , the regime where a mean-based set-point should yield to regime-aware adaptation (cf. Section 8).

*Tier-2 results (DTN-realistic CGR).* We reproduce the findings in dtnsim, a Bundle-Protocol/Contact-Graph- Routing simulator of the CGR lineage, over the same scenarios with a custom energy-harvesting battery module. Result 1: routing periodic status updates over a real contact plan yields a mean delivery delay of 2453.9s against the closed-form  $\mathbb{E}[R] = 2453.9$ s, and the delay histogram reproduces the full mixed residual law (atom plus uniform slab). Result 2: for a two-segment relay, the end-to-end latency standard deviation matches  $\sqrt{\text{Var} R_1 + \text{Var} R_2}$  under incommensurate segment periods (independence) but collapses toward  $\sqrt{\text{Var} R_1}$  under commensurate periods (phase locking), supporting the fixed-plan mixing approximation of Remark 3 and demonstrating its phase-locking failure mode (Lemma 1 remains the exact random-offset result). The renewal residual law (5) holds within 0.5% across contact-gap laws (gamma, and genuinely heavy-tailed Pareto), and the AoI/PAol inversion (Remark 2) appears once  $\text{CV}^2(V) > 1 + 2\mathbb{E}[U]/\mathbb{E}[V]$  per (6). Result 3: we implemented a CGR-native  $k$ -copy router that commits a copy to each of the  $k$  best distinct CGR routes (approximating independence), and compared it against single-copy CGR under random relay faults and a source energy gate. Scenario: the three-relay diamond of Fig. 13; each relay fails and recovers independently with exponential time-to-failure and time-to-repair of mean 8000s and 4000s respectively (each relay is down 1/3 of the time); the source battery gates each copy at cost  $e$  with the harvest-rate sweep setting  $\eta$ ; 10 fault/harvest seeds, 95% CIs; full scenario files ship with the artifact. Two copies cut mean OPAol by roughly  $6\times$  across the entire energy range (Fig. 13; e.g.  $7.0P$  single-



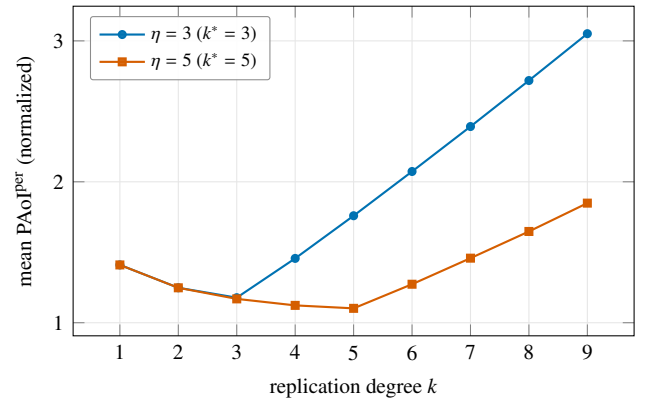
**Fig. 6.** Result 1 (E1/E2): simulated mean AoI and OPAoI vs. duty cycle match the closed forms; the peak grows linearly and the mean quadratically in  $(1 - \delta)$  (provisioning divergence). Markers are Monte-Carlo means with 95% CIs (smaller than markers).



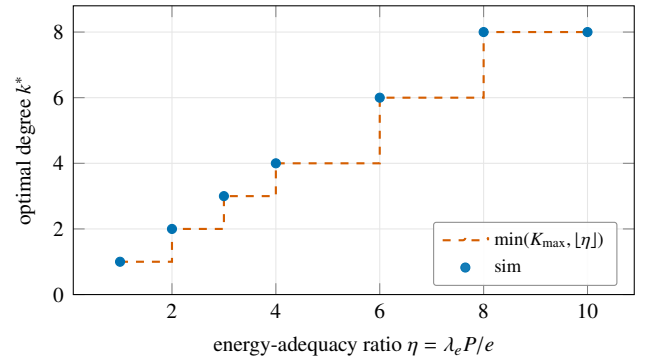
**Fig. 7.** Result 2 (E4): the two-copy gain. Simulated  $\mathbb{E}[R_{\min}]/\mathbb{E}[R]$  matches the two-copy ratio  $\frac{2}{3}(1 - \delta)$ ; replication saves most at sparse contacts ( $\delta \rightarrow 0$ ).

copy versus  $1.2P$  with two copies at  $\eta=6$ , with non-overlapping CIs)—the order statistic over two relay-disjoint CGR routes with independent relay-fault processes hedges relay faults. Because the per-transmission energy gate defers the second copy when the battery is low rather than dropping the update, the benefit *degrades gracefully* as  $\eta$  falls instead of inverting. The threshold’s *starvation* side is then validated in the DTN stack directly with *update-atomic* all-or-nothing admission (launch  $k$  copies iff the battery holds  $k$  units, else skip the whole update) on a fault-free three-relay staggered diamond (Fig. 14): admitted fractions match  $\min(1, \eta G_{\text{gen}}/kP)$  to  $< 0.5\%$  at every  $(\eta, k)$ , and over-replication is punished exactly as (16) predicts—at  $\eta=1$ ,  $k=3$  skips two of three updates and its peak age is  $2.7\times$  single-copy’s. The same experiment measures Remark 8 in real CGR: fault-free, higher degrees *tie* (never strictly win) once  $\eta \geq k$ —the strict benefit needs route-outcome uncertainty.

*Unified experiment: one topology, one admission rule, both metrics.* Because the preceding experiments varied the metric and the admission mechanism together with the question, we close Tier-2 with a single experiment holding everything fixed: the three-relay staggered diamond, update-atomic all-or-nothing admission, relay faults *on* (availability  $2/3$ ), sweeping  $k \in \{1, 2, 3\} \times \eta \in \{1, 2, 3, 6\}$  (10 seeds) and reporting OPAoI, per-update  $\text{PAoI}^{\text{per}}$ , mean delay, delivered/admitted counts, and energy per update ( $= ke$ ) side by side. The two peak metrics agree on every ordering, and the freshness-optimal degree traces  $1 \rightarrow 2 \rightarrow 3 \rightarrow 3$  across  $\eta$ —exactly  $\min(K_{\max}, \lfloor \eta \rfloor)$  in one experiment. The starvation side is large and unambiguous ( $k=3$  at  $\eta=1$ : +144%;  $k=2$ : +71%); the hedging side is directionally consistent but small here ( $-0.6$  to  $-0.8\%$ , within seed noise per cell), and the reason is itself



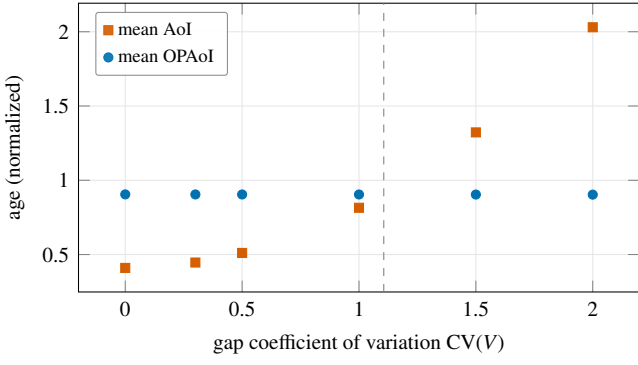
**Fig. 8.** Result 3 (E7): the integrated energy+replication+AoI objective is unimodal in  $k$  with minimum at one of the two integers bracketing  $\eta$ ; pushing  $k$  past the energy budget starves future contacts and raises the peak.



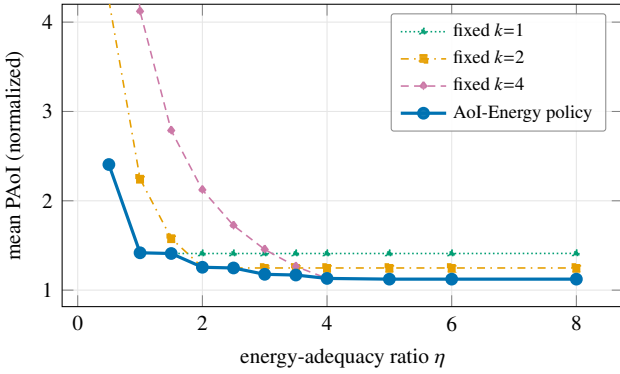
**Fig. 9.** Result 3 (E8):  $k^*(\eta)$  is non-decreasing in the harvest-to-contact ratio and saturates at  $K_{\max}=8$  once energy ceases to bind.

informative: dtnsim’s faults are *stall-type* (a faulted relay pauses, CGR re-routes, no bundle is lost; delivery stays 100%), and in this three-relay staggered plan re-routing recovers most stalls, so the hedge shrinks to residual mid-pass strandings—whereas the route-poor two-relay plan of Fig. 13 converts the same stalls into effective per-pass losses and realizes the large gain. The value of replication is set by the fault-to-loss conversion given the plan’s alternatives—the effective  $q$  of Proposition 1 (far below availability there versus  $\approx 1$  here, Section 6)—a dependence invisible in either experiment alone. Finally, replacing the synthetic LEO schedule with a contact plan generated from real Iridium-NEXT ephemeris (propagated over a polar ground station; cf. polar-coverage constellation design [45]) reproduces the renewal residual law to within 0.2% per satellite (2490.9 s versus the predicted 2492.4 s over 101 passes), confirming the model on operational orbital data.

*Phase alignment on a real constellation.* Extending the ephemeris study to satellite *pairs* instantiates the plan-model taxonomy of Section 6 on real orbits. Sweeping  $\sim 40$  Iridium-NEXT companions of the reference satellite over the same station, the route-set *latency-diversity potential*—the residual-min ratio  $\mathbb{E}[R_{\min,2}]/\mathbb{E}[R_1]$ —spans 0.41 to 0.95 depending on the pair’s pass-phase alignment, tracing the phase ladder (14): an anti-phased pair (residual correlation  $\rho=-0.51$ , plan model (ii)) beats the ensemble reference of  $\frac{2}{3}(1 - \delta)=0.61$ , a near-independent pair ( $\rho=0.07$ , plan model (i)) matches the product form (11) within 4.3%, and a phase-locked pair ( $\rho=+0.88$ , plan model (iii)) collapses the ratio to 0.95. We deliberately do *not* call this a realized replication gain (Remark 8; the  $q=1$  tie below): the scan measures the *diversity replication converts into freshness once route outcomes are uncertain*. Two lessons stand: the dependence of fixed-plan residuals on a common arrival instant is a measured phenomenon on operational constellations, not a technicality; and because contact plans are known in advance, the phase alignment of candidate routes is *computable at planning time* via



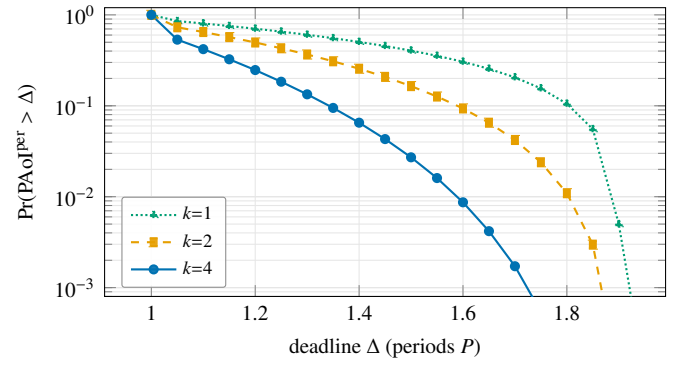
**Fig. 10.** The AoI/OPAoI inversion (E12). Mean OPAoI ( $= T_s + \mathbb{E}[V]$ ) is flat in the gap variability, while mean AoI grows with it; they cross at the exact threshold  $CV^2(V) = 1 + 2\mathbb{E}[U]/\mathbb{E}[V]$  (dashed:  $CV \approx 1.11$  for the plotted  $\mathbb{E}[U]/\mathbb{E}[V] = 1/9$ ), so for high-variance terrestrial contacts the outage peak sits *below* average AoI.



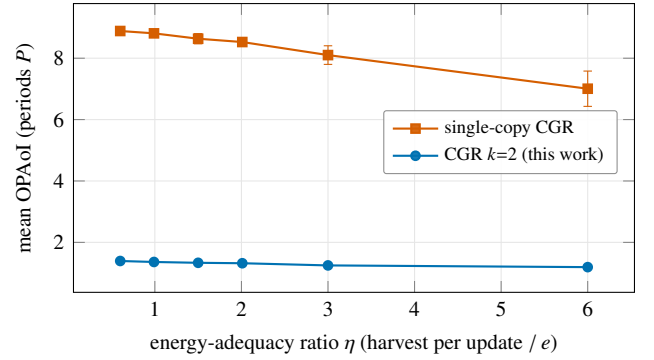
**Fig. 11.** The AoI-Energy policy selects the replication degree from the closed-form rule  $k_{\text{pol}}(\eta) = \arg \min_k \mathcal{A}(k)$  (no search); it matched the per- $\eta$  exhaustive optimum in all 11 cases and is the lower envelope of every fixed-degree curve. A single fixed  $k$  is wasteful when energy is scarce (large  $k$  starves) or stale when it is abundant (small  $k$  misses the order-statistic gain); the adaptive policy is best at every energy level.

(10)—a phase-aware refinement of the route choice in (26).

**Matched-budget baseline comparison.** To place the policy against its routing-family baselines under the *same* copy budget ( $K_{\text{max}}=3$ ), the same battery/harvest process ( $B=12$ , sweep over  $\eta$ ), and the same route selection (all schemes commit copies to the plan-earliest routes; only the *degree* differs), we run the model-exact simulator with the Tier-2 fault duty ( $q=2/3$  per-route survival). Three baselines: earliest-arrival single-copy CGR (in our single-energy-bottleneck abstraction, E-CGR’s battery validation [37] coincides with the energy gate, so the gated  $k=1$  baseline *is* the E-CGR-style policy); RUCoP-style delivery-max replication (energy-blind,  $k=K_{\text{max}}$  always); and the two-candidate  $k^*(\eta)$ . Three regimes emerge (mean  $\text{PAoI}^{\text{per}}$  in units of  $P$ , 10 seeds, 95% CIs below  $\pm 0.04$ ): energy-scarce ( $\eta \leq 1.5$ ),  $k^*=1$  and the policy *equals* single-copy CGR (3.14 at  $\eta=0.5$ ) while delivery-max over-replicates catastrophically (6.47, 2.1 $\times$  worse); transition ( $\eta=2-2.5$ ), the policy strictly dominates both (1.375 vs. 1.640 / 1.785 at  $\eta=2$ ); energy-abundant ( $\eta \geq 3$ ), the policy coincides with full diversity (1.32) while single-copy stays fault-limited (1.64). The same harness re-validates Remark 8 and the phase ladder jointly: on a two-route plan with  $q=1$  (no faults), single-copy CGR *ties*  $k=2$  within CI for every plan structure (locked / independent / anti-phased), whereas at  $q=2/3$  the two-copy set reduces mean  $\text{PAoI}^{\text{per}}$  by 16–20% with the predicted ordering anti (1.378) < independent (1.418) < locked (1.497). Two *battery-state-adaptive* rules complete the baseline set: battery-greedy (fire  $\min(K_{\text{max}}, \lfloor b \rfloor)$  copies each epoch) and battery-threshold (all-in  $K_{\text{max}}$  iff affordable, else a single copy). The mean-based set-point dominates both wherever energy binds—by up to 29% over greedy at  $\eta=1$



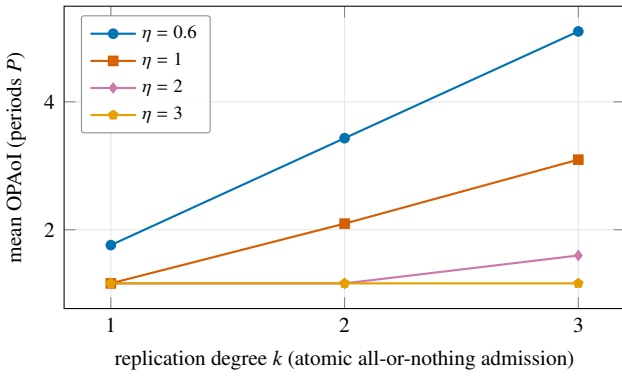
**Fig. 12.** PAAoI tail (deadline-violation probability) at energy-abundant  $\eta$ : replication pulls in the tail via the order statistic. The probability of exceeding a 1.5 $P$  freshness deadline falls from 0.40 ( $k=1$ ) to 0.16 ( $k=2$ ) to 0.027 ( $k=4$ )—a 15 $\times$  reduction—and the 99th-percentile PAAoI drops from 1.90 $P$  to 1.59 $P$ . Mean PAAoI alone understates the benefit of replication for mission-critical traffic.



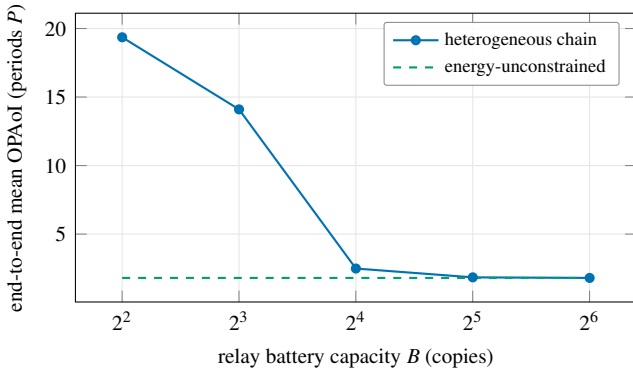
**Fig. 13.** Replication benefit in a real CGR stack with our CGR-native  $k$ -copy router (dtnsim, random relay faults, source energy gate; 10 seeds, 95% CIs). Sending two copies over the two best distinct CGR routes cuts mean OPAoI by roughly 6 $\times$  versus single-copy across the whole energy range—the order statistic hedges relay faults. Because the per-transmission energy gate defers the second copy when the battery is low (rather than dropping the update), the benefit degrades gracefully as  $\eta$  falls instead of inverting. This experiment isolates the fault-hedging *mechanism*; it is not a test of the atomic all-or-nothing policy of (17), whose over-replication penalty appears in Fig. 14.

(1.70 vs. 2.20), where adaptive rules over-spend scarce harvest—while in the energy-abundant regime greedy edges the set-point by 2.6% at  $\eta=3$  (1.286 vs. 1.321): with faults present, firing a *partial* degree beats the occasional all-or-nothing skip. The all-or-nothing restriction, not the two-candidate degree, is what greedy exploits there; relaxing it is a natural refinement. The uncertainty-aware objective itself is validated model-exactly (E18): simulated mean  $\text{PAoI}^{\text{per}}$  matches (21) within 0.7% over  $q \in \{0.4, 0.7, 1\} \times k \leq 4$  at  $\eta \in \{3, 6\}$ , with the optimum in the two-candidate set throughout. Scenario and outputs ship with the artifact (`e_baselines.py`, `e18_uncertain.py`).

**Reference-implementation cross-check (ION).** As a final tier we reproduce the Result 1 anchor in ION 4.1.4, the reference Bundle Protocol implementation [31]: two ION nodes (BPv7, ipn scheme) exchange status bundles over a contact plan that opens a  $W=6$  s window once per  $P=60$  s (the period compressed so that 120 cycles fit in two hours of wall-clock time, since ION runs in real time). Over 974 bundles sent at a 7.3 s cadence (incommensurate with  $P$ , so arrival phases quasi-uniform), delivery is lossless and the delay histogram reproduces the mixed residual law—atom near zero plus a uniform slab over the 54 s gap. A per-bundle regression of the measured delay on the plan-predicted residual gives slope 1.016—the residual law holds one-for-one per bundle—with the intercept and the gap-arrival lag ( $2.50 \pm 0.51$  s, near-constant) identifying ION’s contact-activation/dispatch latency: precisely the lumped service term  $T_s$  of the model, here measured in the reference implementation (the realized-phase residual mean is



**Fig. 14.** Tier-2 atomic threshold test (real CGR, fault-free 3-relay staggered diamond, update-atomic admission: launch  $k$  copies iff the battery holds  $k$  units, else skip the whole update; 5 seeds). The over-replication penalty is now visible in the DTN stack: at  $\eta=1$ ,  $k=3$  admits 35% of updates (theory  $\min(1, \eta G_{\text{gen}}/kP)$ , matched to  $< 0.5\%$ ) and its peak age is  $2.7\times$  single-copy's. At  $\eta \geq 3$  the degrees tie exactly: with a fault-free deterministic plan, oracle CGR's single copy already rides the earliest route, so extra copies deliver stale duplicates—the strict replication benefit requires route-outcome uncertainty (Fig. 13).



**Fig. 15.** Energy coupling in the full heterogeneous chain (sensor  $\rightarrow$  UAV mule  $\rightarrow$  LEO  $\rightarrow$  gateway in real CGR). A finite battery at the UAV/LEO relays must cover the per-contact forwarding burst; below that, end-to-end mean OPAoI degrades by  $\sim 11\times$  (and delivery collapses from 96% to 31%). With sufficient battery it matches the energy-unconstrained chain, whose mean delay confirms the additive composition  $\mathbb{E}[Y]=\mathbb{E}[R_1]+\mathbb{E}[R_2]+\mathbb{E}[R_3]$  to 1.1%.

24.34 s versus the 24.30 s closed form). We stress this is a reference-implementation cross-check on a Linux host, not a flight test; its value is that ION has flight heritage. At operational periods ( $P \approx 5.7 \times 10^3$  s) this  $T_s$  is  $\sim 0.04\%$  of the mean: the gating law, not the software overhead, governs freshness.

**Full heterogeneous end-to-end scenario.** To exercise the complete stack we run the canonical chain sensor  $\rightarrow$  UAV mule  $\rightarrow$  LEO  $\rightarrow$  gateway in dtnsim, each hop carrying its own contact model: high-variance (gamma,  $\text{CV}=1.3$ ; over-dispersed but light-tailed) terrestrial gaps, a periodic-with-jitter UAV segment, and a deterministic LEO segment, with incommensurate cadences. Across ten seeded replications (each with its own realized contact plan and per-seed additive prediction  $\mathbb{E}[R_1]+\mathbb{E}[R_2]+\mathbb{E}[R_3]$ ), the simulated mean end-to-end delay matches the prediction to within 5.9% on every seed and to 0.2% in the across-seed mean (7165 versus 7176 s), supporting the fixed-plan mixing approximation of Remark 3—and the additive composition it licenses—across heterogeneous segment types in a real CGR stack; e.g. for the reference seed, 6754 s measured against  $3617+758+2454=6829$  s predicted over a three-hop path with 96% delivered, full-chain mean OPAoI  $10.3 \times 10^3$  s. Adding a finite battery at the UAV/LEO relays exposes the energy coupling in the multi-segment setting (Fig. 15): a relay battery must cover the per-contact forwarding burst, and below that threshold end-to-end PAoI degrades by an order of magnitude.

*Where the low-load regime ends.* The uncongested-contact scope of Section 3 is quantified by loading contacts to capacity: we fix the window's carrying capacity at  $m$  bundles (contact rate set so a 430 s window serializes and volume-books exactly  $m$ ; CGR is volume-aware) and sweep the offered update rate from  $\sim 1$  to 16 per period at  $k=1$  (single relay) and  $k=2$  (staggered diamond; each relay window sees  $\sim 2/3$  of the copies). Below capacity the low-load model is sharp—mean delay stays within 12% of the unloaded value all the way up to *and including* the offered-to-capacity point  $\rho_{\text{load}}=1$  (copies per window equal to  $m$ ; paced deterministic arrivals are stable at capacity), and OPAoI keeps *improving* with offered load as more updates ride each window—then fails as a step, not a knee: at  $\rho_{\text{load}}=2$  delay inflates  $119\times$  and delivered fraction drops to exactly the capacity ratio. Replication doubles the offered contact volume, so  $k=2$  reaches the same wall at half the update rate ( $\rho_{\text{load}}=1.33$  at 16 updates/period with  $m=8$ ; delay  $\times 98$ ). The boundary of validity is thus sharp and predictable—offered copies per window below  $m$ —and inside it the freshness analysis stands; beyond it, contact-volume management (overbooking, CGR-ETO [36]), not freshness-optimal replication, is the binding problem.

## 10. Conclusion

We studied information freshness over scheduled, intermittently connected multi-segment DTNs with energy-constrained bundle replication. One alternating-renewal model yields the residual-wait laws for LEO, UAV, and terrestrial segments and the exact inversion condition (6) under which peak age falls below average age; phase mixing (exact under random offsets, measured on fixed plans) makes end-to-end latency compose, with the two-copy gain equal to half the dispersion of path latencies. For the periodic gating bottleneck, the energy budget admits a near-threshold optimal degree ( $\lfloor \lambda_e P / e \rfloor$  or the skip-aware neighbor, capped at  $K_{\text{max}}$ ; monotone and saturating), extended by Proposition 1 to per-copy route success  $q$ —so route-outcome uncertainty, the true source of the replication gain, enters the same objective, and replication under uncertainty buys *cadence*, not only latency spread. The floor/ceiling structure is proved below the floor and holds beyond it under an explicit, grid-verified increment condition; finite battery exerts downward pressure on the optimum (a downward shift in the tested i.i.d.-harvest scenarios).

All analytical results are numerically verified by model-exact Monte-Carlo; representative results—the residual laws, the atomic starvation penalty, and the fault-hedging benefit—are corroborated in a DTN-realistic Bundle-Protocol/CGR simulator, and the residual-wait anchor is independently reproduced in the ION reference implementation (lossless delivery, the law exact per bundle, and the dispatch overhead measurable as the model's  $T_s$ ); the threshold further survives bursty (Markov-modulated) harvest up to a mean dwell of ten contact periods, and real Iridium-NEXT ephemeris confirms both the residual law per satellite and the pair-phase dependence of the route-set latency-diversity potential—suggesting *phase-aware copy placement* on anti-phased contacts as a planning-time refinement. The framework extends, rather than replaces, the CGR/RUCoP contact-plan stack, so the energy-aware replication degree can be applied on top of an existing route computation; a comparison against a RUCoP-style MDP core on an illustrative instance shows the energy-aware degree cap attains most of the MDP's delivery at a fraction of the energy. Ongoing work treats regime-aware degree adaptation under strongly persistent harvest (the measured dwell-50 boundary—at the network edge, a continual-learning task [62]), Markov-modulated contact schedules beyond the renewal assumption, and integrating RUCoP's route set under the energy-aware degree cap.

## Data availability

Code, contact-plan scenarios, all result data (CSV), and the scripts that regenerate every figure are publicly available at <https://github.com>.



com/UNO-Networks-Lab/paoi-scheduled-dtn (MIT license) and archived on Zenodo (doi:10.5281/zenodo.21841317); the project page mirrors the manuscript and slides.

### CRedit authorship contribution statement

**Haoyu Wang:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Bo Sheng:** Conceptualization, Methodology, Supervision, Validation, Writing – review & editing. **Xiaoqian Zhang:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing, Funding acquisition.

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### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI tools (ChatGPT; Claude) to assist with language refinement, manuscript organization, literature cross-checking, and critical review of drafts. The authors independently verified the mathematical derivations, experimental results, references, and all scientific claims; reviewed and edited the resulting text; and take full responsibility for the content of the article.

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